Tectonophysics 811 (2021) 228872

Contents lists available at ScienceDirect

Tectonophysics

journal homepage: www.elsevier.com/locate/tecto

Paleomagnetic results from Upper Triassic and Middle Jurassic strata of east-central New Mexico, and implication for North American apparent polar wander path

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ARTICLE INFO

Keywords:
Paleomagnetism
Inclination error
North American mid-Jurassic Apparent Polar Wander Path
Elongation/Inclination
Summerville Formation
East-central New Mexico

ABSTRACT

Two apparent polar wander (APWP) paths for North America (NA) diverge during the Jurassic. The path based on paleomagnetic results from igneous rocks of eastern NA, well-dated Ontario kimberlites, and carbonates and radiolarites of Adria (northern Italy - transformed to NA coordinates) has a large gap at high latitude, which could document rapid plate motion or True Polar Wander. The APWP from coeval NA sedimentary rocks define a lower-latitude, more continuous path. The discrepancy between the two paths has been attributed to inclination error (IE) from deposition and compaction of the sediments.

New paleomagnetic results from Upper Triassic (Garita Creek, Trujillo, and Redonda Formations) and Middle Jurassic (Entrada, Summerville, and Lower Morrison Formations) sedimentary units of east-central New Mexico have positive reversal and fold tests, which indicate their magnetization, mostly carried by detrital hematite, is primary. Interpretation of noisy and non-linear demagnetization paths was facilitated by a smoothing scheme, which did not bias the mean directions. Using sample-level directions from line fits, IE was estimated ~20° for the Jurassic units, but is negligible for the Triassic units because mean directions are horizontal. Mean paleomagnetic poles for the ~223 Ma (Garita Creek and Trujillo) and ~214 Ma Redonda Formations are 55.7°N/78.7°E/A95 = 3.2° and 59.7°N/81.0°E/A95 = 4.0°; those for the ~163 Ma Summerville and ~155 Ma Morrison Formations are 59.3°N/138.1°E/A95 = 6.5° and 48.7°N/151.8°E/A95 = 7.8°.

Simulated ~163 Ma paleomagnetic poles based on Summerville results and a wide range of IE show that no amount of IE correction to the Summerville directions would produce poles consistent with the high-latitude Jurassic APWP. Furthermore, paleomagnetic poles for Trujillo, Redonda, Summerville, and Lower Morrison Formations of east-central New Mexico, even with IE-correction, streak approximately along the 60° parallel (present coordinates) for most of late Triassic and Jurassic, which validates the lower-latitude APWP. Another explanation for the APWP discrepancy is required.

1. Introduction

Combinations of rapid plate motion and the role and occurrence of True Polar Wander (TPW) (e.g., during Jurassic) have been proposed as explanations for several large-scale geologic and tectonic processes (e.g., Torsvik et al., 2008, 2012; Kent et al., 2015; Muttoni and Kent, 2019; Fu et al., 2020). The primary evidence for these plate motions and possible TWP are derived from analysis of paleomagnetic apparent polar wander paths (APWPs), many of which are composites of paleomagnetic studies from separate plates that are combined after correction for differential motion between individual plates. The advantages of these composite, or imported, APWPs, is that larger sets of results from better recorders of the geomagnetic field, and from more precisely dated rock units, can be used to assess plate motion on a global basis (e.g., Torsvik et al., 2008). They also, serve as a cratonal reference path for an individual plate (e.g., Enkin, 2006; Kent and Irving, 2010). Of course, the accuracy of these imported APWPs depends on accuracy of the plate reconstructions used, their appropriateness for the terrane from which the poles were derived, and errors associated with them that are generally not propagated accordingly. One important element in these APWP compilations is the selection and evaluation of data and data quality; this is particularly important for hematite-bearing sedimentary rocks whose record of

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https://doi.org/10.1016/j.tecto.2021.228872
Received 31 August 2020; Received in revised form 8 April 2021; Accepted 12 April 2021
Available online 19 April 2021
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paleomagnetic inclination may be influenced by inclination error (which will be referred to as IE in this paper). The currently accepted practice for hematite-bearing rocks is to use paleomagnetic results from studies where the effects of IE have been evaluated and corrected for, if necessary, using a mineral-alignment based method (e.g., Kodama, 1997; Tan and Kodama, 2003), analysis of the shape (ellipticity) of large sets of directional data (e.g., Tauxe et al., 2008; Tauxe and Kent, 2004), or an assumed degree of IE is inferred (e.g., Torsvik et al., 2012).

Current efforts to better define the Jurassic APWP produced paleomagnetic results from well-dated kimberlites. Those yielded paleomagnetic poles with high geographic latitudes that require rapid, northerly motion of North America between ~160 and ~ 145 Ma (Kent et al., 2015; Kent and Irving, 2010). Some paleomagnetic poles from units of other plates, rotated to North American coordinates for comparison, also have high geographic latitudes that are consistent with the Jurassic kimberlite results (Fu et al., 2020; Kent and Irving, 2010; Muttoni and Kent, 2019). The rapid pace of apparent polar wander along this high-latitude APWP and hence motion of North America (and other plates) has been used to propose an episode of rapid plate motion referred to as the Jurassic "monster shift" and/or TPW between ~160 Ma and ~145 Ma (Kent et al., 2015; Muttoni and Kent, 2019). There are, however, a noticeable number of paleomagnetic results from sedimentary rocks within North America that are coeval with this proposed large, rapid shift in the motion of North America. In contrast to the results that support the high-latitude APWP, their paleomagnetic poles fall at lower latitudes, and define a different APWP. In both Kent and Irving (2010) and Kent et al. (2015), the difference between the APWP defined by these Jurassic sedimentary rocks and the high-latitude APWP for North America recorded by the kimberlites was attributed to paleomagnetic IE for the sedimentary rocks, so those results were omitted from the Kent and Irving (2010) and Kent et al. (2015) compilations. These different paths, and the resulting discussion of the paleomagnetic results used to generate those paths, mirrors debate in the literature regarding the North American Jurassic APWP several decades ago (e.g., Hagstrum, 1993).

Until recently, there had been few direct tests for the effects of and corrections for IE on Jurassic sedimentary rocks of the American southwest. Muttoni and Kent (2019) reported a thorough study of exposures of the Kimmeridgian-Tithonian Morrison Formation from western Colorado (USA), which was previously studied by Steiner and Helsley (1975). The discussion and background for this problem, and the paleomagnetic results from the Morrison Formation (Bazard and Butler, 1994; Steiner and Helsley, 1975; Van Fossen and Kent, 1992), are extensively reviewed and summarized by Muttoni and Kent (2019), so the reader is referred to that paper for further detail. The conclusion of their work was that the majority of samples from the Morrison Formation had a pervasive overprint, which Muttoni and Kent (2019) tentatively attributed to deep weathering in early Cretaceous time, and that the characteristic directions that most resemble those found by Steiner and Helsley (1975) and Bazard and Butler (1994) was scattered, has normal and reverse polarity directions that are not antipodal, and whose paleomagnetic pole positions did not match either those from previous work, or from the results that defined the high-latitude North America APWP. These observations led Muttoni and Kent (2019) to conclude that the complexity of the paleomagnetic results from the Morrison Formation precludes their use in APWP compilations.

The paleomagnetism of Triassic - Jurassic Formations on the Colorado Plateau has been extensively studied (e.g., Bazard and Butler, 1992, Bazard and Butler, 1991; Bazard, 1991; Steiner and Helsley, 1974; and from CO-drilling program- see Kent et al., 2019 and references in that). However, paleomagnetic results from correlative units farther east, such as in central and eastern parts of New Mexico, are sparse and mostly from Upper Triassic strata (Molina-Garza et al., 1995, 1996, 1998; Reeve and Helsley, 1972; Zeigler and Geissman, 2011), with few results from some of the Middle Jurassic units (Steiner, 2003; Steiner et al., 1994). None of these studies sought to evaluate possible effects of IE, and/or the number of existing observed directions is too small to permit a robust examination of IE using methods such as the Elongation/Inclination technique of Tauxe and Kent (2004).

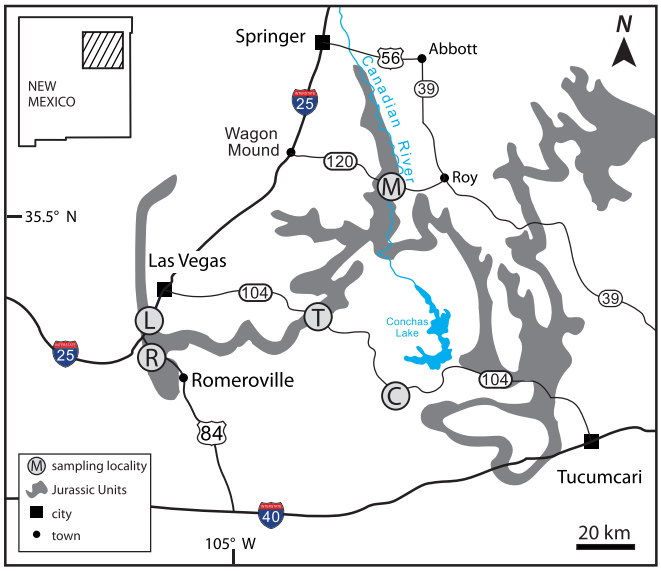


Fig. 1. Sampling localities in east-central New Mexico; R: Romeroville Exit, L: Las Vegas city, T: Trujillo Mound, C: Conchas Lake, M: Mills Canyon; The gray shading shows the distribution of Middle through Upper Jurassic strata [modified from Anderson and Lucas, 1992].

The upper Callovian-Oxfordian Summerville Formation provides an opportunity for an additional test of the effects of IE on the paleomagnetic signal of these hematite-bearing sediments. The ages of these strata (165–158 Ma) overlap the proposed transition from a polar-wander stillstand from 190 to 160 Ma to rapid plate motion or TPW proposed by Muttoni and Kent (2019), and are comparable in age to the Ontario Kimberlites (156 Ma) (Kent et al., 2015). As was the Morrison Formation, the Summerville Formation has been studied previously, with the most recent and complete results from Bazard and Butler (1992), and Steiner (2003). It likely preserves a remanent magnetization from sedimentation or shortly after deposition (Steiner, 2003). The sections sampled by Steiner (2003) are located in east-central New Mexico, and are attractive for this purpose in that they are not located on the Colorado Plateau, and thus will not be affected by rotation of the Plateau. Therefore, this formation was targeted for extensive sampling (Fig. 1).

Here we present paleomagnetic results from Upper Triassic and Middle Jurassic formations of east-central New Mexico, which enabled us to propose magnetostratigraphy and paleomagnetic poles of the studied sections. Moreover, obtaining paleomagnetic directions from the Summerville Formation at two localities with different amounts of tilting, one near the city of Las Vegas, NM and recently exposed layers at Mills Canyon [Fig. 1]), allowed us to perform a conclusive fold test to evaluate the age and fidelity of these magnetization directions. Finally, we will discuss the new Middle Jurassic results in comparison with the previous works and also discuss, through simulations, whether IE caused the North American APWP discord.

2. Geology

Triassic and Jurassic sedimentary rocks of the U.S. Southwest interior are dominantly shallow marine carbonates, fluvial sandstones and mudrocks, evaporites, and eolianites that accumulated in shallow marine to non-marine depositional environments (DeCelles, 2004). In east-central New Mexico, Triassic units encompass the Chinle Group, which begins with trough-crossbedded quartzarenites of the Santa Rosa Formation (Lucas et al., 2001). The overlying Garita Creek Formation, dominated by red-bed mudstones, is in turn overlain by the Trujillo

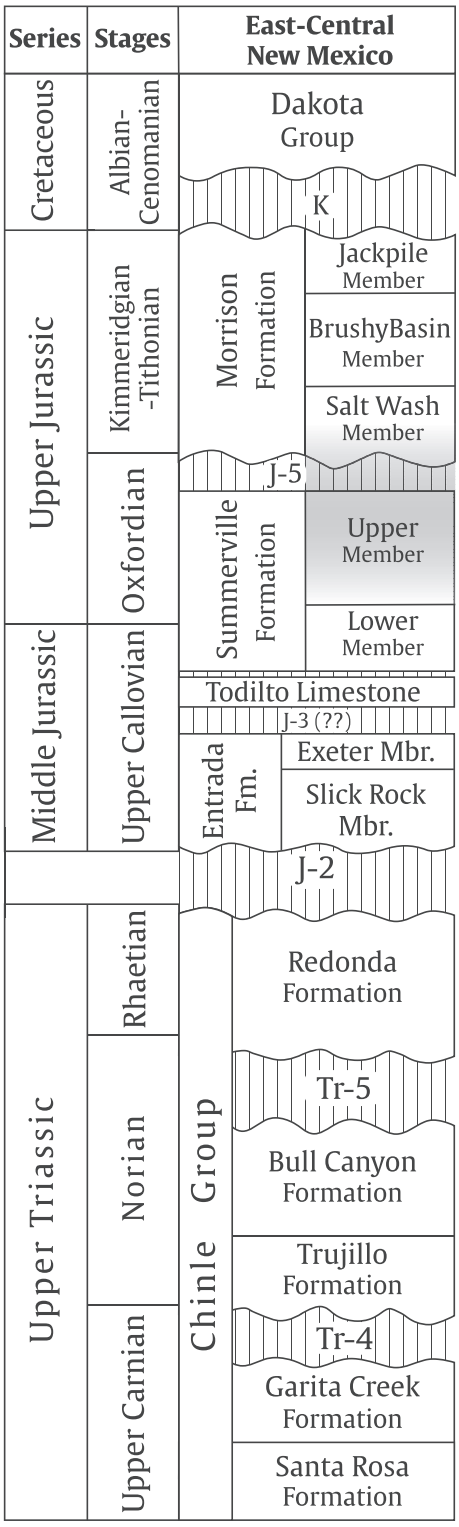


Fig. 2. Stratigraphy of Upper Triassic to Cretaceous strata in east-central New Mexico. Sources for stratigraphic columns are Lucas (2004) and Lucas et al. (2001) for Upper Triassic, Lucas (2018), Lucas and Woodward (2001), and Anderson and Lucas (1992) for Jurassic, and Lucas and Anderson (1998) for Cretaceous. The shaded unit was interpreted as Lower Morrison Formation in the Las Vegas section (see Fig. 1) by Steiner et al. (1994).

Formation, which consists of trough-crossbedded micaceous litharenites with thin interbeds of red-bed mudstone. Red-bed mudstones of the Bull Canyon Formation rest on top of the Trujillo Formation, and they are overlain by cyclic beds of reddish-brown siltstone and sandstone of the Redonda Formation, the youngest formation of the Chinle Group (Lucas, 1995) (Fig. 2). Disconformably on top of the Redonda Formation, Jurassic units begin with trough-crossbedded yellowish-gray to pale reddish-brown eolian sandstone of the Slick Rock Member of the Entrada Formation (Lucas and Woodward, 2001) (Fig. 2). Dark-gray limestone of the Todilto Formation rests disconformably on the Entrada Formation. The disconformably overlying Summerville Formation has lower and upper parts. The lower part consists of cyclically bedded sandstone and mudstone, whereas the upper part is mudstone and siltstone (Lucas and Anderson, 1997). Sandstones and conglomerates of the Salt Wash Member of the Morrison Formation disconformably overlie the Summerville Formation (Anderson and Lucas, 1996; Lucas and Anderson, 1998) (Fig. 2).

Most of these lithostratigraphic units are bounded by unconformities (Lucas and Spielmann, 2013) (Fig. 2). Due to the similarity of the lithologies, lateral discontinuity of the unconformities, and existence of local (and some unknown) disconformities (Lucas, 1995; Lucas et al., 1999; Pipiringos and O'Sullivan, 1978), sample sites were grouped into sets organized by geographic location and relative stratigraphic position of each site within that location's section.

2.1. Localities and their sampled sets

Upper Triassic and Middle to Upper Jurassic units were sampled from roadcuts at five localities: *Romeroville Exit* from I-25 (route 84), along I-25 near *Las Vegas (NM)*, on route 104 at *Trujillo Mound* ~ 50 km to the east of the Las Vegas, farther to the southeast on the same road near *Conchas Lake*, and much farther to the north on route 120 at *Mills Canyon* (Fig. 1). At each locality, the sample sets, based on lithology, were named by letters indicating the location (e.g., R for Romeroville Exit, and M for Mills Canyon), followed by another letter referring to stratigraphic position in ascending order. More information about the sampled units is reported in Table 1. There we include unit names used by the workers cited above, but note that refinements or re-interpretations of the local stratigraphy can be applied to the new paleomagnetic data using the detailed sample locations provided.

2.1.1. Romeroville Exit

Sets RA, RB, and RC, from a 35 m section (including sampling gaps) sampled on the eastern side of road 84, Romeroville Exit from Interstate-25 (Fig. 1), appear to be from the Garita Creek and Trujillo Formations in the middle of the Chinle Group, and thus late-Triassic (Carnian - Norian) in age (personal communication, Dr. S.G Lucas, 2019).

RA set is dominantly red shale and siltstone with fine sandstones interbeds and includes about three meters of maroon, moderately-bedded fine sandstone with thin white bands of crossbedding, and conglomerate lenses at the base. These properties match the lower shale member of the Chinle Formation (now Group) described by Kelley (1972), which later was raised to formation rank and called Garita Creek Formation by Lucas and Hunt (1989). This set is disconformably (Tr-4 unconformity) overlain by the RB & RC sets.

RB and RC sets: the RB set is thin-bedded grayish-tan medium-grained sandstone with a layer of conglomerate at its base. Above are the thick-bedded sandstones of the RC set. These are the Cuervo Sandstone member of (Kelley, 1972), which are now called the Trujillo Formation of the Chinle Group (Lucas, 1995; Lucas and Hunt, 1989).

Table 1
Sets and zone divisions at each sampled locality. Zones are 'N' and 'R' for normal and reverse polarity zones, respectively.

Set	Zone	Coordinates		Lithostratigraphic unit	Bedding attitude		Lithology	Apprx. thickness (m)	Apprx. sampling spacing (m)
		Lat.	Long.		Dip direction (°)	Dip (°)			
Trujillo Mound									
TD	TD-R1	35.519	255.307	Morrison Fm./Salt Wash Sst	Horizontal		creamy sst	10	1
TC	TC-N1	35.516	255.310	Entrada Fm./Slick Rock Mbr.	060	06	tan sst, with x-bd (eolianite)	7	1.5
	TC-A1[a]	35.516	255.310	Entrada Fm./Slick Rock Mbr.	060	06		1.5	1.5
TB	TB-R	35.514	255.316	Redonda Fm.	327	03	red sst. & mdst.	27	0.4–0.5
TA	H02 (TA-N)	35.514	255.316	Redonda Fm.	Horizontal		white ls.	1	0.4–0.5
	H01 (TA-R)	35.514	255.316	Redonda Fm.	[b]			1	0.4–0.5
Mills Canyon									
MC	MC-R1	35.912	255.650	Morrison Fm.	310	04	pale gray sst.	1–2	1
	MC-N1	35.912	255.650	Morrison Fm.	310	04		3–4	1
MB	MB-R1	35.913	255.653	Summerville Fm.	324	04	red sltst. & mdst./ with alabaster nod.	8	0.4
	MB-N1	35.913	255.653	Summerville Fm.	302	05		0.5	0.4
MA	MA-A2[a]	35.914	255.652	Entrada Fm./Exeter Mbr.	280	04	mud-support sst.	1	1
	MA-A1[a]	35.914	255.652	Entrada Fm./Exeter Mbr.	280	04		5	1
Las Vegas									
LA	LA-R2	35.525	254.759	Morrison Fm./Salt Wash Sst.	074	30	creamy sst.	6	1–2
	LA-N2	35.525	254.759	Summerville Fm.	067	32	pale red sst. & sltst. with sh. & mdst.	15	1–2
	LA-A2[a]	35.525	254.759	Summerville Fm.	074	30		3	1–2
	LA-R1	35.525	254.759	Summerville Fm.	073	26		20	1–2
	LA-A1[a]	35.525	254.759	Summerville Fm.	074	30		2	1–2
	LA-N1	35.525	254.759	Summerville Fm.	078	32		6	1–2
	H01 (LA)	35.525	254.759	Todilto Fm.	074	30	dark gray ls.	1–2	0.4–0.5
Romeroville exit									
RC	RC-N	35.517	254.755	Trujillo Fm.	065	12	creamy sst.	8	0.5–1
RB	RB-N	35.517	254.755	Trujillo Fm.	073	15	t.b. sst.	4	0.3
RA	RA-N	35.517	254.755	Garita Creek Fm.	076	12	red sst. & sltst.	3	0.1–0.15
Conchas Lake									
CB	CB-A1[a]	35.298	255.708	Trujillo Fm.	268	02	creamy sst. with large x-bd	6	1
	CB-N1	35.298	255.708	Trujillo Fm.	268	02		8	1
CA	CA-N1	35.298	255.708	Trujillo Fm.	355	08	red sltst.	1.5	0.2

[a] Parts of sections in which a group of stratigraphically consecutive samples have atypical directions compared to the mean direction of their adjacent polarity zones (most differ more than 40°). These parts are not included in final calculations.
[b] Discarded from further analysis due to initially undetected local slumping in the sampled horizon. lat. and long.: coordinates of the sampling sites (datum: WGS84); sst: sandstone; sltst: siltstone; mdst: mudstone; ls: limestone; sh: shale; x-bd: cross-bedding; t.d.: thin-bedded; alt: altered; nod: nodule; Fm: Formation; Mbr: member.

2.1.2. Las Vegas

Approximately 200 m of Jurassic strata are exposed at Romeroville Gap, a roadcut on I-25 near Las Vegas city (Fig. 1). The section encompasses Todilto, Summerville, and the Morrison Formations (e.g., Lucas and Woodward, 2001). Two sampled sets (LA, and LB) are separated by an interval covered by Quaternary deposits. The LA set (~55 m) is dominantly pale-reddish sandstone and siltstone, whereas LB is grayish-green thin-bedded mudstone. The latter, which resembles the Brushy Basin member of the Morrison Formation and is overlain by Mesa Rica sandstone of the Dakota Group, did not yield interpretable results and was omitted from further analysis.

The first two meters of the LA set consist of dark-green organic-rich limestone of the Todilto Formation (Lucas, 2018; Lucas et al., 1999; Lucas and Woodward, 2001). The limestone set is overlain by ~10 m of red mudstone with interbedded sandstone layers of Summerville Formation (Anderson and Lucas, 1992). Above this, there are ~40 m of pale-red thick-bedded sandstone with some dark laminae, which was previously assigned to the Salt Wash member of Morrison Formation (Steiner, 2003; Steiner et al., 1994) (shaded-area in Fig. 2). However, further stratigraphic analysis has shown that this sandstone-dominated portion should be included as part of the Upper Summerville Formation because it is homotaxial with the Bluff sandstone of the San Rafael Group (Lucas et al., 1999; Lucas and Woodward, 2001). The LA set is

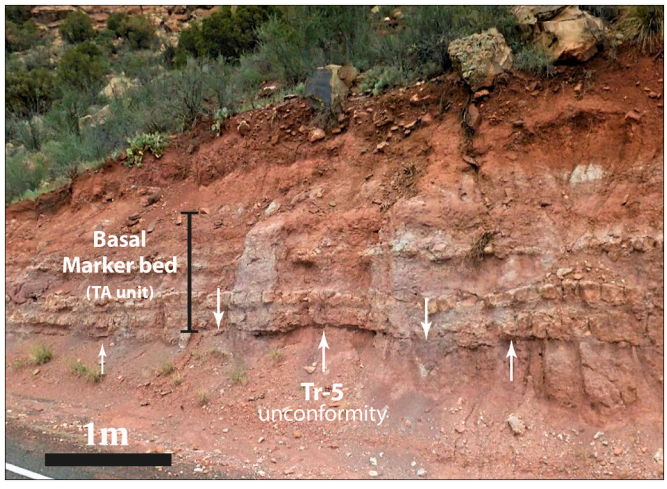


Fig. 3. TA limestone unit, which is the Basal Marker bed (Hester and Lucas, 2001) of Redonda Formation of Upper Chinle Group. Arrows point to the Tr-5 unconformity.

unconformably (J-5) overlain by the Salt Wash Sandstone of Morrison Formation, which consists of yellowish-brown, coarse-grained, trough-crossbedded sandstone containing some clay clasts (Lucas et al., 1999), and here is as thin as about 5 m (Lucas, 2018) (Fig. 2).

The Todilto Formation equivalent to the northwest in Utah, the Curtis Formation, contains early Oxfordian ammonites (Imlay, 1980). Considering this and the age of the overlying Morrison Formation, deposition of Summerville Formation was during Oxfordian time (ca. 163–157 Ma, according to the time scale of Walker et al., 2018) (Wilcox and Currie, 2008).

2.1.3. Trujillo Mound

At Trujillo Mound, ~50 km east of Las Vegas city on route 104 (Fig. 1), the road cuts through Mesozoic strata from the Upper Triassic Chinle Group, which forms most of the valleys and low-lands, to the Cretaceous Dakota sandstone, which holds up mesas around the area. Sampled units are Upper Chinle Group, Redonda Formation (TA and TB sets), Exeter sandstone (TC set), Salt Wash sandstone of the Morrison Formation (TD set), and Mesa Rica sandstone (TE set). The last one did not yield interpretable results so was discarded from further analysis.

TA and TB sets: About two meters of light-gray limestone layers, lying disconformably on underlying strata, make up the TA set (Fig. 3). The silty, bioturbated, and micritic nature of this limestone along with its stratigraphic position identify it as the "basal marker bed" of the Redonda Formation (Hester and Lucas, 2001) (Fig. 3).

Over the basal marker bed, there are ~27 m of red, fine- to medium-grained sandstone and interbedded mudstone and siltstone of the TB set. Characteristics such as low-angle trough-cross bedding and bedding thicknesses resemble the "SI" facies of Hester and Lucas (2001) of the Redonda Formation. The age of Redonda Formation is late-Norian to Rhaetian, determined by correlation of the Apachean land-vertebrate faunachron (LVF) (Lucas and Hunt, 1993), which is pinned by the first appearance of a tetrapod genus "*Redondasaurus*" (ca. 215 Ma) to the standard global chronostratigraphic scale (Lucas, 2010).

TC and TD sets: About seven meters of well-sorted tan sandstones of the TC set rests unconformably on of the Redonda Formation. Its characteristics match the Slick Rock Eolianites of the Entrada Formation (e.g., Lucas and Woodward, 2001). Exposures of the TC set are traceable all around the escarpments in this area. About 20 m of thin-bedded mudstone of Summerville Formation overlie the TC set, but are largely obscured by debris so was not sampled. This mudstone is disconformably overlain by ~10 m of thick-bedded, creamy-white channel sandstone of the TD set, which resembles the Salt Wash sandstone of the Morrison Formation.

2.1.4. Conchas Lake

About 30 km to the southeast of Trujillo Mound on route 104, near the Conchas Dam Lake, there is another exposure of the Trujillo Formation (Fig. 1) (personal communication, Dr. S.G Lucas, 2019). This section consists of ~14 m of creamy-white to yellowish-tan, coarse grained, large trough-crossbedded sandstone of the CB set, which is underlain by about one and half meters of brownish-red mudstone layers of the CA set.

2.1.5. Mills Canyon

Mills Canyon, carved by the Canadian River, exposes Middle to Upper Jurassic strata (Wanek, 1962) (Fig. 1). Five lithological sets (MA, MB, MC, MD, and ME) were sampled in this section. Unfortunately, MD and ME sets did not yield interpretable results.

MA set: Lowermost set of this section consists of about seven meters of loosely consolidated, coarse-grained, pebbly sandstone of Entrada Formation (Exeter Member). Most samples yielded very noisy paleomagnetic results.

MB set: The MA set is gradationally overlain by about nine meters of freshly exposed alternating pale-red fine sandstone and mudstone. Alabaster nodules are abundant in the sandstone layers. These layers are lenticular and commonly pinch out in the outcrops. These are typical characteristics of the unit formerly known in the area as the Bell Ranch Formation (Wanek, 1962), which has been shown to be the eastern extent of the Summerville Formation of the Colorado Plateau (Lucas et al., 1999; Lucas and Woodward, 2001). At the time that Mankin (1972) studied the stratigraphy of this section, the Morrison Formation was mapped as directly overlying the Entrada Sandstone. This was because the Summerville strata were not fully exposed - the lenticular nature of this unit caused it to pinch out across the sections. Road reconstruction in 2011 formed the new roadcuts that uncovered these beds.

MC set: The MB set is overlain by about six meters of pale-gray, coarse-grained, and medium-bedded sandstone of the MC set. The lithology matches the Salt Wash sandstone of the Morrison Formation, although the J-5 unconformity of Pipiringos and O'Sullivan (1978), which separates the Summerville Formation from its overlying unit (Fig. 2), was not found.

3. Sample acquisition, preparation, measurement and rock magnetic analysis

3.1. Acquisition, preparation, measurement and demagnetization

All samples were obtained as oriented cores in the field using standard methods. In the Pacific Northwest Paleomagnetism Laboratory at the Western Washington University, the 2.5 cm in diameters cores were sliced into specimens 2.2 cm in length. Remanent magnetization was measured with a 2G Enterprise model 755 three-axis DC-SQUID magnetometer in a magnetically shielded room, which has an internal field of less than 350 nT. Progressive thermal demagnetization was employed to reveal components of Natural Remanent Magnetization (NRM) of specimens using the ASC model TD48 furnace. Progressive alternating field (a.f.) demagnetization used an ASC D-2000 unit with peak fields of 200 mT.

A total of 345 specimens, 190 from Upper Triassic and 155 from Middle Jurassic formations, were demagnetized. Samples that were suspected to have magnetite as their main remanence carrier were demagnetized using a.f., which was generally ineffective, and at best a large portion of NRM (generally >50%) remained after a.f. demagnetization. At least one specimen per layer was selected for pilot thermal demagnetization with temperature steps of 50–100 °C below 500 °C, 10–30 °C between 500° and 630° or 650 °C, and 3–10 °C above 650 °C. After this revealed the general demagnetization behavior of each set, most of the specimens were thermally cleaned at larger intervals below 500 °C, 30–40 °C steps up to 600° or 630 °C, and very small steps (3–10 °C) for temperatures above 630 °C.

In most samples, it appeared that hematite is the main carrier of the remanence, and unblocking temperature ranges above 650 °C were used to reveal the Characteristic Remanent Magnetization (ChRM) component of specimens. Linear demagnetization paths (at high unblocking temperature ranges) were quantified with Principal Component Analysis (PCA) of Kirschvink (1980), while the great-circle (GC) fits (McFadden and McElhinny, 1988) were employed for non-linear paths. To not bias directions by possible relatively high-temperature overprints, unanchored line-fits were used for most specimens. For those few specimens whose linear path clearly trended toward the origin, line-fits were anchored to the origin. Mean paleomagnetic directions for zones with only line-fits were calculated by Fisher (1953) statistics; for zones with combined line and GC fits the method of McFadden and McElhinny (1988) was used. Data analysis was carried out using PuffinPlot v. 1.03 (Lurcock and Wilson, 2012).

3.2. IRM

Isothermal remanent magnetization (IRM) acquisition was measured using a MicroMag 3900 Princeton Measurements Corporation vibrating

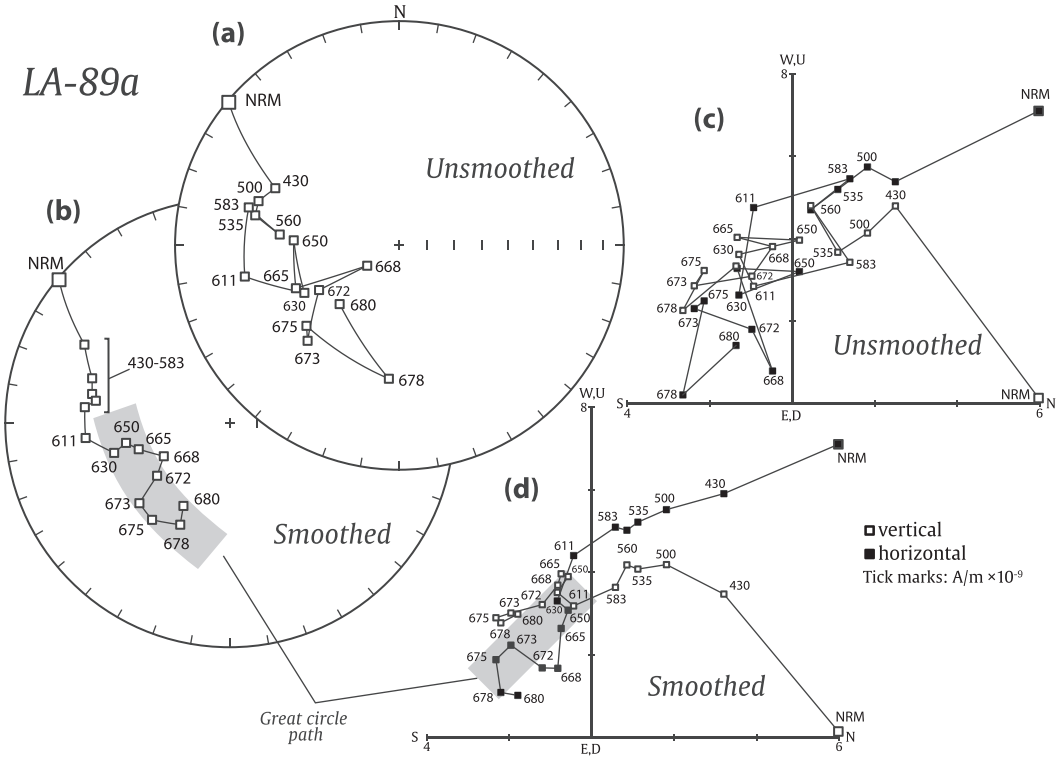


Fig. 4. Equal-area and orthogonal vector endpoint diagrams (before tilt correction) of raw (a, and c) and smoothed (b, and d) data for a representative thermally demagnetized sample from the LA-R2 zone. The contrast shows the effectiveness of the averaging scheme to decrease noise in the data. Note that the great-circle path of remanence vectors (650–680 °C, shaded) was not as perceptible before smoothing.

sample magnetometer (VSM). Most of the samples contained a high-coercivity phase (e.g., goethite and/or hematite) that did not saturate in the 2.2 T maximum field of the VSM. Remanence of the same samples was measured with DC-SQUID magnetometer after progressive stepwise magnetization up to 5.0 T produced with an ASC impulse magnetizer. To combine the IRM measurements from the two methods we overlapped three measurements at 1500, 1850, and 2200 mT. The average differences of these steps were used to adjust the results from the impulse magnetizer method to match those of from the VSM.

Assuming that the IRM acquisition curves follow a cumulative log-Gaussian (CLG) function, the curves were decomposed by the method of Kruiver et al. (2001) to identify magnetic mineralogy of samples. For example, magnetite would have a low mean coercivity ($B_{1/2}$) compared to hematite, and a population with broad grain size range would have a larger distribution (DP) than one with a narrow size range.

4. Results, tests, and their interpretation

4.1. Data pre-processing

4.1.1. Averaging (smoothing)

One of the main issues with data interpretation was that demagnetization paths for many specimens were not clearly linear or non-linear due to the high level of noise. This became more problematic at higher temperatures, where signal-to-noise ratio was small during unblocking of most of the ChRM components above ~650 °C. To reduce the visual impact of the noise (Fig. 4), we decided to smooth (average) measurements using a 0.5–1.0-0.5 weighting scheme. This means that for each treatment step we used the following: [(0.5 × previous measurement) + (the measurement) + (0.5 × the following step)] / 2; and for the last step to be consistent with the rest of the modified steps, used (0.5 × previous + last step)/1.5.

We acknowledge that perhaps the major concern with smoothing is that it smears the temperature-moment relationships. However, in practice, the uncertainty of the temperature gradient in the oven (~10 °C) would have already disturbed the temperature-moment relationship. To minimize contribution of smoothing to the uncertainties, we did not fit lines (or GCs) where temperature steps were more than 10 °C apart.

To determine the ChRM direction of specimens, lines, or GCs (Fig. 4), were fitted on the smoothed version of data and then the same T_{ub} range and fit type were used on the raw data. The maximum angular deviations (MADs) were generally <10° for the smoothed data, but larger (20°-30°) over the same intervals of the raw data. The zone-mean paleomagnetic directions of raw and smoothed datasets are very similar with generally better precision for the smoothed version (see Table 2), which shows that smoothing did not bias the results and was beneficial to identify the ChRMs as it increased signal to noise ratio.

4.1.2. Site division

Initially, the two or three samples from each sampled horizon were grouped as a site in the hope that each would provide a good average of the field at time of deposition. However, because most of the specimens had very noisy and non-linear demagnetization paths, the best approach seemed to be to group samples from adjacent horizons that preserved a single polarity zone. Each group defined in this way we call a 'zone' (Table 1). Thus, zone-mean directions come from a group of specimens from adjacent horizons that appeared to belong to a single polarity interval.

4.2. Upper Triassic Formations

Upper Triassic units are Garita Creek, Trujillo, and Redonda Formations. Except for the limestone layers of the TA set (basal marker bed [Fig. 3]) of the Redonda Formation, the rest of the Triassic strata are terrigenous rocks. NRM intensities range between 1.24×10^{-4} and 2.96

Table 2
Paleomagnetic directions of Upper Triassic and Middle Jurassic strata from east-central New Mexico. n/N: number of directions used to calculate the zone mean/ total number of specimens; Dec and Inc.: tilt-corrected declination and inclination of the characteristic remanent magnetization (ChRM) component; a95 and k: half-angle of the 95% confidence limit and precision parameter of Fisher (1953); R: length of the resultant vector; L/GC: number of line fits (principal component analysis of Kirschvink, 1980) / great-circle fits used to calculate the mean (McFadden and McElhinny, 1988); Lat. (Long.): north latitude (east longitude) of the paleopole (negative sign indicates south latitude).

Set	Zone	n/N	L /GC	Raw data							Smoothed data						
				Mean direction					VGP		Mean direction					VGP	
				Dec	Inc	a95	k	R	Lat.	Long.	Dec	Inc	a95	k	R	Lat.	Long.
Trujillo Mound																	
TD	TD-R1	21/41	20/1	129.4	−36.9	11.5	8.63	18.74	−43.5	340.7	128.4	−37.5	11.1	9.18	18.88	−42.8	341.8
TC	TC-N1	9/13	9/–	348.4	49.1	17.0	10.17	8.21	78.8	139.0	350.0	43.0	15.9	11.50	8.30	76.4	117.4
	TC-A1	5/5	5/–	053.1	56.2	28.6	8.09	4.51	47.6	327.3	053.4	53.1	29.5	7.67	4.48	46.6	331.7
TB	TB-R	78/112	45/33	176.6	−04.3	7.2	5.79	67.56	−56.5	261.5	177.2	−04.2	6.6	6.71	68.98	−56.5	260.5
TA	H02 (TA-N)	3/9	2/1	355.0	14.3	14.5	100.37	2.99	61.4	85.8	355.4	14.1	13.7	112.95	2.99	61.3	084.8
	H01 (TA-R)	5/5	5/–	155.5	−36.2	16.5	22.36	4.82	−63.6	316.2	152.8	−33.4	9.1	71.21	4.94	−60.4	316.8
Mills Canyon																	
MC	MC-R1	4/9	2/2	132.1	02.8	57.7	4.2	3.53	−31.9	316.6	159.8	1.5	22.7	22.9	3.91	−48.8	287.2
	MC-N1	5/7	5/–	314.8	41.6	28.5	8.2	4.51	49.4	161.3	314.9	41.9	27.8	8.5	4.53	49.6	161.6
MB	MB-R1	24/26	7/17	153.6	−18.2	10.7	8.6	22.31	−54.2	304.2	151.3	−16.3	11.0	8.1	22.21	−52.0	306.2
	MB-N1	4/5	3/1	343.1	32.0	25.5	15.5	3.84	66.1	119.1	343.6	29.8	20.8	22.9	3.89	65.3	116.1
MA	MA-A2	3/5	1/2	212.8	17.2	–	3.2	2.68	−35.7	214.4	250.3	22.0	91.1	6.5	2.85	−8.7	186.6
	MA-A1	12/13	8/4	100.8	−13.5	28.4	3.2	9.19	−12.7	344.9	103.5	−10.9	32.3	2.7	8.63	−14.2	342.2
Las Vegas																	
LA	LA-R2	5/6	1/4	146.6	−44.6	24.4	16.0	4.87	−60.1	336.1	150.9	−36.6	21.2	20.9	4.90	−60.3	321.8
	LA-N2	7/22	6/1	313.5	42.4	19.1	11.2	6.51	48.7	162.5	310.7	37.9	19.1	11.2	6.51	44.8	160.1
	LA-A2	4/6	1/3	82.3	−33.9	47.3	7.8	3.81	−04.7	004.3	091.0	−42.7	30.2	18.0	3.92	−14.9	004.8
	LA-R1	21/28	10/11	152.8	−15.8	15.3	5.1	18.19	−53.0	303.5	151.5	−17.5	13.5	6.5	18.75	−52.8	303.1
	LA-A1	6/6	6/–	283.9	73.3	46.2	3.1	4.37	36.8	216.1	294.0	72.4	48.3	2.9	4.26	41.9	213.6
	LA-N1	8/10	8/–	330.5	39.3	19.7	8.9	7.21	61.1	145.4	331.3	36.4	20.0	8.7	7.19	60.5	141.3
	H01 (LA)	2/3	1/1	159.2	−18.4	–	7.5	1.93	−57.8	295.8	154.7	−18.8	–	5.5	1.91	−55.3	302.6
Romeroville exit																	
RC	RC-N	21/58	21/–	001.9	05.8	13.1	6.8	18.08	57.3	071.2	001.3	03.6	13.0	7.0	18.12	56.3	072.4
RB	RB-N	9/26	9/–	350.4	05.7	20.9	7.0	7.86	56.2	092.1	351.7	8.1	20.3	7.4	7.92	57.6	090.4
RA	RA-N	23/64	22/1	003.7	−04.5	8.2	14.7	21.54	52.1	068.7	004.7	−3.5	6.1	25.2	22.15	52.5	067.1
Conchas Lake																	
CB	CB-A1	4/17	1/3	050.0	−53.4	33.7	14.6	3.90	06.5	035.9	049.5	−56.7	36.3	12.7	3.88	04.1	038.4
	CB-N1	13/35	12/1	350.2	−00.3	13.6	10.2	11.88	53.4	092.4	348.7	−00.4	15.1	8.5	11.65	53.0	094.7
CA	CA-N1	9/18	9/–	351.7	03.3	7.0	55.3	8.86	55.4	090.5	351.8	03.1	6.8	59.1	8.86	55.4	090.3

$\times 10^{-2}$ A/m. Demagnetization behavior of the representative specimens of the Upper Triassic units along with either the same or a sister specimen's IRM decomposition results (Kruiver et al., 2001) are presented in Fig. 5.

All of the Carnian-Norian (ca. 223 Ma) age Garita Creek and Trujillo Formations' zone-mean directions have north seeking and shallow magnetizations interpreted as normal polarity (Fig. 5a-e). However, there is a zone identified at the top of the Conchas Lake section (CB-A1) that had an east-directed and moderately-steep mean direction, unlike those expected for the late Triassic. That could be due to magnetization during an excursion and/or transition period or incomplete removal of the overprint signal. After removal of a (viscous) low-temperature component from many specimens at 150°-330 °C, NRM intensities gradually decrease at middle temperature ranges between 250–400 °C and 580–620 °C, which could be due to unblocking of magnetite and/or pigmentary hematite. The stable ChRM components were isolated between 660 °C and 685 °C.

Paleomagnetic directions of the Redonda Formation (ca. 214 Ma) preserved two polarity reversals. Although the first layer (zone TA-R) of the TA set was discarded from magnetic pole calculations, its magnetization was acquired in a reverse polarity. In the TA set, after removal of a low temperature (150–200 °C) component, the NRMs gradually decay on an almost linear path toward the origin between ~300° and 675 °C with a slightly sharp drop after 640 °C (Fig. 5f). This is from unblocking hematite with a wide grain size distribution; no low-coercivity (<100 mT) constituent (e.g., magnetite) exists in this set, also confirmed by its IRM decomposition results (Fig. 5f).

In many specimens, the IRM acquisition curves were decomposed into three magnetic phases (Fig. 5). The phase normally with the highest $B_{1/2}$ (mean coercivity), generally >1000 mT, could be iron oxyhydroxide (e.g., goethite), as there are detectable drops of NRM intensities at 150–200 °C. The soft fraction ($B_{1/2}$ < 100 mT) contributions to the IRM in the Garita Creek and the Trujillo Formations are between 25 and 40%, while in the Redonda specimens they are <10% (0% in the TA set). Generally, the dominant constituent, and likely the remanence carrier, is in the high-coercive fraction. In the Trujillo Formation this dominant phase has an average mean coercivity ($B_{1/2}$) of ~225 mT and dispersion parameter (DP) of ~0.31 mT, whereas in the Redonda Formation the mean $B_{1/2}$ and DP are ~550 and ~0.37 mT, respectively. This difference may indicate a coarser grain size of the ferromagnetic population in the Trujillo Formation than the Redonda Formation.

Both the remanence decay curves and the IRM decomposition analysis show that the hard fraction is mostly hematite, which in some sets (e.g., CB and TB sets [Fig. 5e, g-h]) has distinctively different population of grain sizes. It is likely that pigmentary hematite corresponds to the smaller grain-size population, as the DP values <0.30 mT also hint for a narrow grain-size distribution and thus a possible chemical source. Because the unblocking temperature (T_{ub}) range of pigmentary hematite is usually <500–600 °C (Jiang et al., 2015), which overlaps with magnetite's T_{ub} range, ChRM components were isolated at higher temperature ranges to avoid contributions from pigmentary hematite and magnetite.

4.3. Middle Jurassic Formations

Sampled Middle Jurassic units are Todilto, Entrada, Summerville,

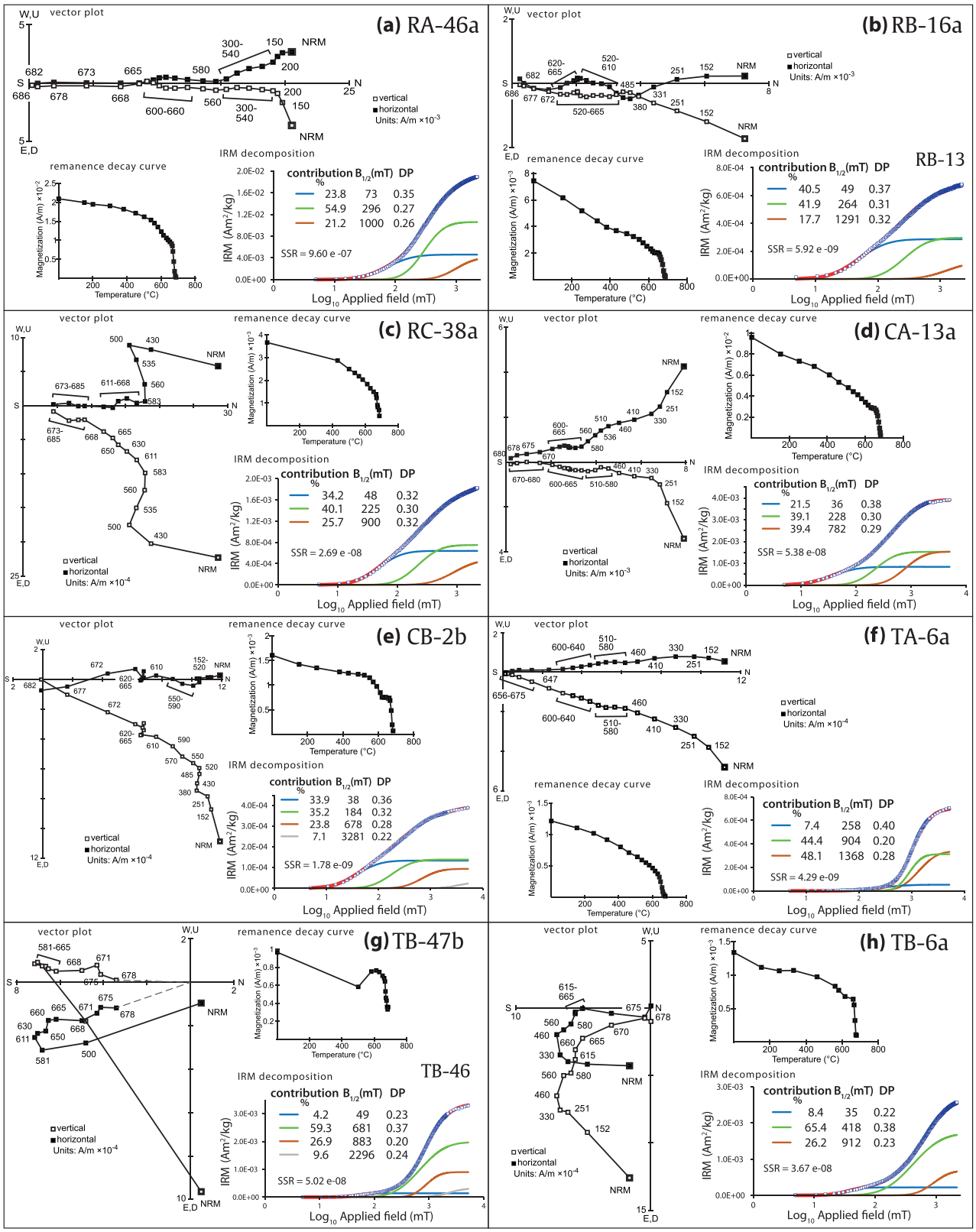


Fig. 5. Orthogonal vector endpoint diagrams and remanence decay curves of thermally demagnetized samples representative of Upper Triassic zones, with their isothermal remanent magnetization (IRM) plots (log-acquisition plot). (a-b, d-e) for zones of the Trujillo Formation (CA-N1, CB-N1, RB-N, and RC-N respectively). (c) for the RA-N zone of the Garita Creek Formation. (f) for the TA-N zone and (g-h) for the TB-R zone of the Redonda Formation. IRM decomposition results based on cumulative log-Gaussian (GLC) function (Kruiver et al., 2001). $B_{1/2}$ and DP are the mean coercivity and dispersion parameter of the GLC function. SSR is the sum of the squared residuals for IRM component fitting.

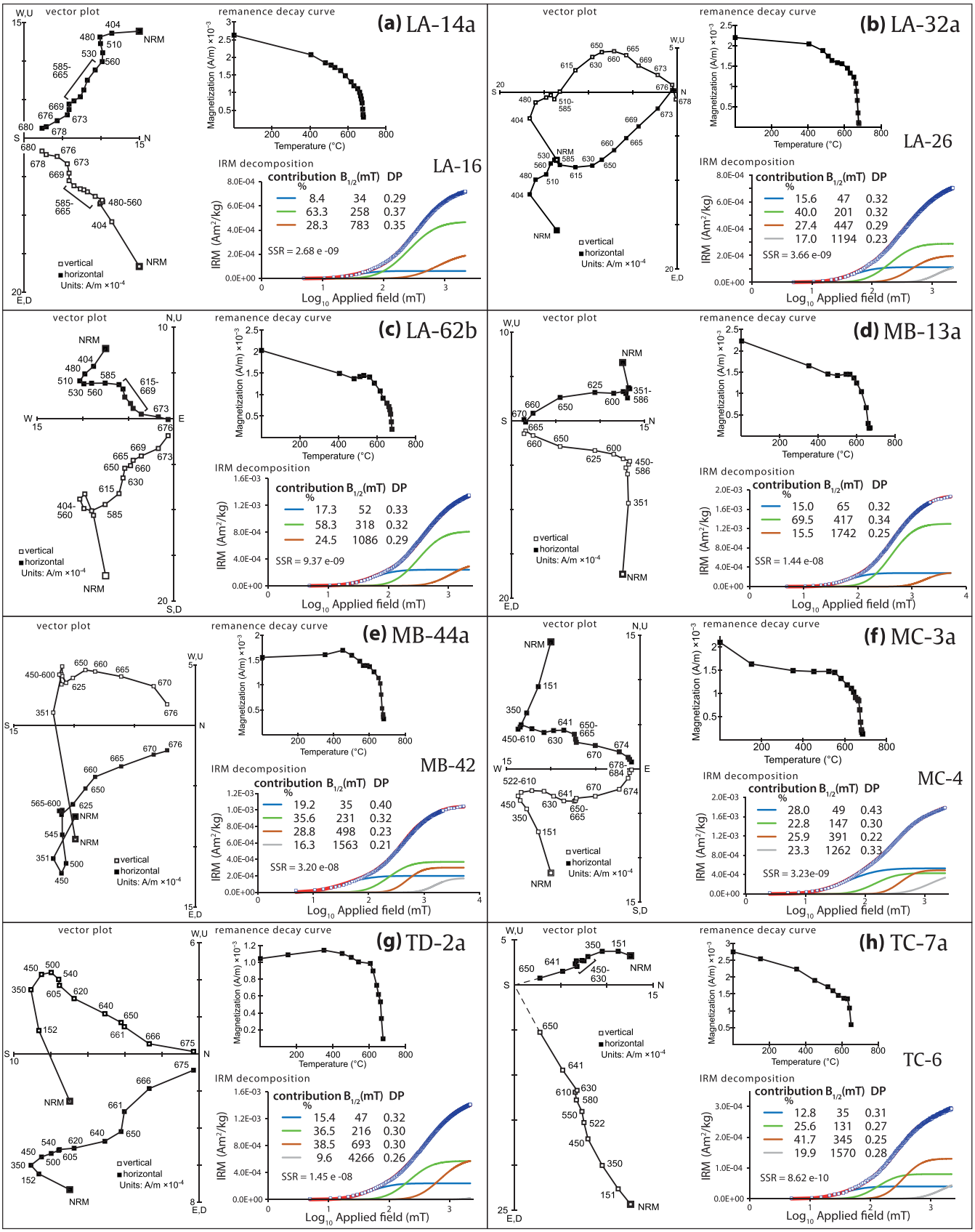


Fig. 6. Orthogonal vector endpoint diagrams and remanence decay curves of thermally demagnetized samples representative of Middle Jurassic zones, with their IRM plots (log-acquisition plot). (a-e) for zones of the Summerville Formation (LA-N1, LA-R1, LA-N2, MB-N1, and MB-R1 respectively). (f, h) for the MC-N1 and TD-R1 zones of the Morrison Formation. (g) for the TC-N1 zone of the Entrada Formation. See Fig. 5 for more information.

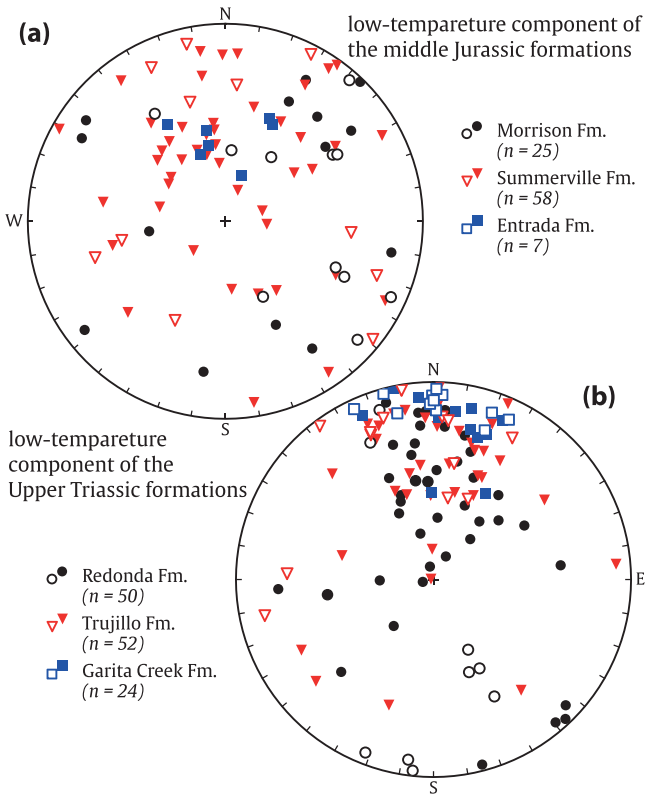


Fig. 7. Low-temperature component directions (downward [solid] and upward [open]) from free line fits with MAD < 20° isolated between ~200–300 °C and ~400–550 °C observed in (a) Middle Jurassic formations: Entrada Fm. (squares), Summerville Fm. (triangles), and Salt Wash mbr. of Morrison Fm. (circles); and (b) Upper Triassic formations: Garita Creek Fm. (squares), Trujillo Fm. (triangles), and Redonda Fm. (circles) of east-central New Mexico. Unlike the "B" component from Muttoni and Kent (2019) [their Figure 4] the low-temperature component in these units is scattered so the overprint is unlikely to bias ChRM of Middle Jurassic and Upper Triassic strata of the east-central New Mexico, and the stable ChRM at much higher temperature ranges (>650 °C) could be isolated and interpreted as significant.

and Morrison Formations (Table 1). There is only one zone in the Todilto Formation. Although it recorded a reverse polarity, its results are not well-defined enough to be included in later analysis. More than 90% of the mid-Jurassic samples have an NRM intensity range between 1.93×10^{-5} and 3.18×10^{-3} A/m. Demagnetization paths of the representative specimens of the Middle Jurassic units along with either the same or the sister specimen's IRM decomposition results (Kruiver et al., 2001) are presented in Fig. 6.

Slick Rock Sandstone of the Entrada Formation was only sampled at Trujillo Mound, right above the J-2 unconformity. Paleomagnetic directions from the bottommost part of this set showed large deviations from the directions found in adjacent strata. The rest of this set preserved a moderately steep north seeking direction interpreted as normal polarity (Table 2). The stable ChRM components were isolated between 630 °C and 650–690 °C (Fig. 6h).

The late Carnian-Oxfordian age (ca. 163 Ma) Summerville Formation was sampled at Mills Canyon and Las Vegas localities (Table 1 and Fig. 1). At both sections, paleomagnetic reversals were recorded. During thermal demagnetization NRM intensities decreased 10–30% by 350° to 500 °C in most of the specimens (Fig. 6a-e). A coherent and discernable component was not observed over the expected T_{ub} range of magnetite. In many specimens, a component demagnetized between ~590° and 660 °C was different in direction than the stable ChRM component with T_{ub} range from 660° to 680 °C, which is clearly evident in Fig. 6b. Our inference is that the former could be carried by the pigmentary hematite, while the latter is from detrital hematite. In final calculations of the paleomagnetic pole of the Middle Jurassic Formations, only the zone-means of the reversed polarities were averaged. This is because the next to last component removed (~590° to 660 °C, mentioned above) was generally easier to identify in the reversely magnetized specimens (Fig. 6).

Only the lower parts of the Morrison Formation were sampled as the focus of the project was on the Summerville Formation. Generally, a (viscous-) low-temperature component was removed at 350°/500 °C (Fig. 6f-g). The ChRM component started to unblock gradually after 600°/ 620 °C, with a sharp drop between 660° and 685 °C (Fig. 6f) in some specimens.

Our results differ from those of Muttoni and Kent (2019), who revisited Morrison Formation at the classic section in Norwood, Colorado, previously studied by Steiner and Helsley (1975). They reported that both lower and upper parts of the Morrison Formation carried a coherent (low-) temperature component with a steeply down to the north direction (Figure 4b of Muttoni and Kent, 2019), referred there as the "B" component, and suggested it was from deep weathering during Cretaceous time. Data for their B component includes NRMs and (very) low-temperature steps (~100 °C) for >80% of line fits (see Muttoni and Kent (2019) supporting information 2, Table S3). Incorporation of recent overprints seems likely as the mean direction of this component is very close to the present-day GAD field at Norwood. No such component could be isolated in the demagnetization results of the Upper Triassic and Middle Jurassic Formations of east-central New Mexico, which is in marked contrast with their results. Instead, free line fits with MAD < 20 over temperature ranges 150–350 °C to 400–500 °C (Fig. 7) are very scattered within all units. This T_{ub} range is much lower than for the ChRMs which is mostly isolated above 650 °C. Therefore, the effect of any low temperature component here is negligible.

The IRM acquisition decomposition analysis revealed three to four magnetic phases (Fig. 6). Although a phase with the $B_{1/2} > 1000$ mT was identified in many specimens, it might not be goethite as a significant decrease in NRM intensities around 150 °C was not observed. The soft fraction ($B_{1/2} < 100$ mT) contributions to the IRM in the Entrada and the Summerville Formations are <20% and in the Morrison Formation is 15–30%. The dominant constituent of the IRM has high mean coercivities in all three Formations. This phase has an average mean coercivities ($B_{1/2}$) of ~345 mT, ~260 mT, and ~ 540 mT in the Entrada, the Summerville, and the Morrison Formations, respectively. Except for the Entrada Formation the dispersion parameter (DP) of the dominant constituent for all other Formations is >0.30 mT. The differences in the averaged mean coercivities may reflect differences in the grain size of the corresponding ferromagnetic mineral.

In the Summerville Formation samples, the contribution of a soft fraction to the IRM acquisition curves is small and in agreement with the small decrease of NRM intensities in the T_{ub} range expected for magnetite. However, the soft fraction in the Morrison Formation is less likely to be in magnetite than in maghemite. This is because demagnetization over the T_{ub} range of magnetite was not observed, and NRM decrease at 350°/500 °C might be due to the inversion of maghemite (γFe_2O_3) to hematite (αFe_2O_3) (Dunlop and Özdemir, 1997).

Like the Upper Triassic units, both the remanence decay curves and the IRM decomposition analysis of the Middle Jurassic units indicate that the hard fraction is mostly hematite. Also, the main remanence may reside in that fraction. In the Summerville Formation, it appears that two population of hematite coexist, one with average fine grain size and narrow distribution (~ $B_{1/2}$ = 470 mT, DP = 0.26 mT), and the other coarser with wider distribution (~ $B_{1/2}$ = 260 mT, DP = 0.33 mT) (Fig. 6b and e). The latter could correspond to detrital hematite, while the former likely is the pigment. Similarly, in the Morrison Formation two populations of hematite grains are observed. Although the fine-grain distribution is the dominant component in the IRM, the ChRM components were isolated at high T_{ub} ranges (> 660 °C) in agreement with the coarser grain-size distribution (Fig. 6g-h).

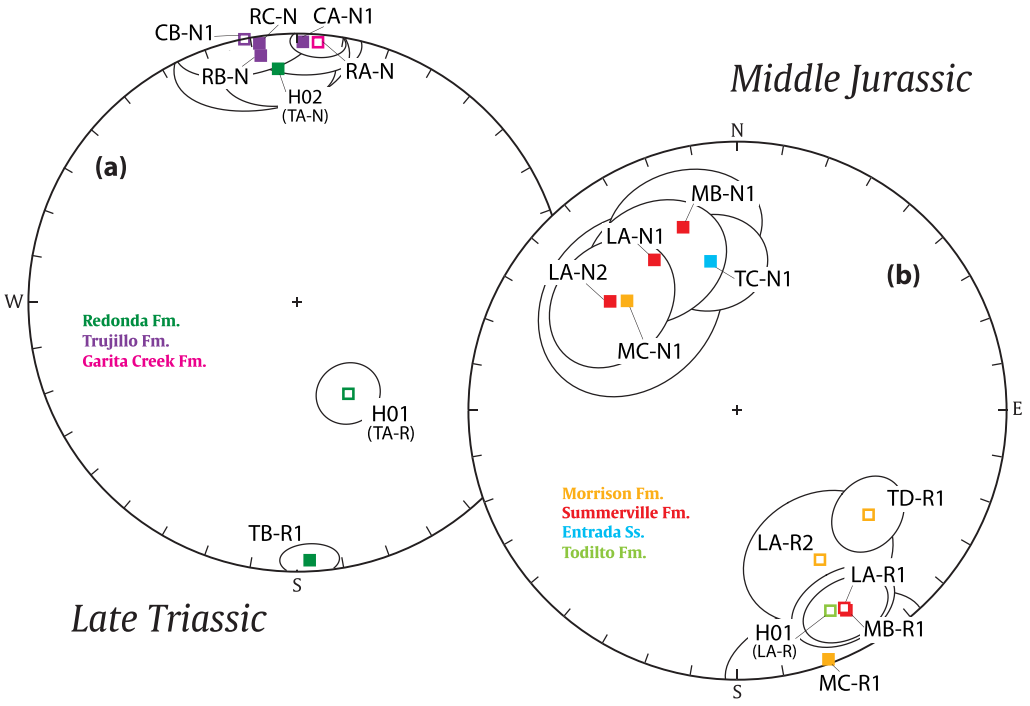


Fig. 8. Equal-area plots of zone-mean directions of (a) Upper Triassic and (b) Middle Jurassic units. Ellipses are the projection of the cone of 95% confidence about the mean directions (not drawn if >30°). TA-R mean direction is discarded from further analysis due to initially undetected local slumping in the sampled horizon.

Paleomagnetic directions from the sampled Upper Triassic and Middle Jurassic strata are reported in Table 2. Results are from component fitting on both raw and the smoothed versions of data. The zone-mean directions, calculated from line fits (PCA of Kirschvink (1980)) with MAD <10° (< 5° for anchored lines), and line fits combined with great-circle fits (McFadden and McElhinny, 1988) with MAD <10° are presented in the Supplementary Tables 1 and 2, respectively. The zone-mean paleomagnetic directions of raw and smoothed datasets are very similar with generally better precision for the smoothed version (see Table 2), which shows that the smoothing did not bias the results and was beneficial to identify the ChRMs as it increased signal to noise ratio. Therefore, for our further analysis and discussions, we preferred to use the smoothed paleomagnetic directional data presented in Table 2. Paleomagnetic mean directions for zones that constitute the polarity zones (Table 2) are shown in Fig. 8.

4.4. Stability tests

4.4.1. Fold test

In east-central New Mexico, most of the sedimentary strata are nearly horizontal (dips <10°) (Johnson, 1974). An exception is near the Romeroville Gap, where uplift of the Sangre de Cristo Mountains during the Laramide orogeny tilted strata ~30° – 40° (Skotnicki, 2003). Acquiring enough paleomagnetic mean directions from the Summerville Formation at both flat (Mills Canyon) and tilted (Las Vegas) sections of this monocline allowed us to examine the stability of the direction through the fold test using the bootstrap technique of Tauxe and Watson (1994).

The test results are shown in Fig. 9. Five zone-mean directions, two from the Mills Canyon and three from the Las Vegas section, were used for the parametric bootstrap test. Although the results show tighter clustering in the tilt-corrected coordinates (Fig. 9a), the range of tilt correction required to maximize the largest eigenvalue (τ_1) was too broad for the fold test to be conclusive. This issue is probably due to asymmetry of the fold and/or IE variation across the units that raises the chance of failure of the Tauxe and Watson's (1994) test with small-sized datasets (McFadden, 1998). Effects of small, local rotations may also be possible.

To overcome this issue, simulated mean directions and their statistical parameters were calculated for each zone. Synthetic datasets were made using each zone's statistical parameters, then their mean directions were calculated. The number of iterations used to generate synthetic zone-mean directions was equal to the number of directions in each zone. The results of the 10 k bootstraps (Fig. 9b-c) indicate a positive fold test at the 95% confidence limit with maximum τ_1 between 73 and 139% (Fig. 9b). This suggests that the magnetization was acquired before folding.

4.4.2. Reversal test

As several polarity reversals were observed in the Upper Triassic and the Middle Jurassic strata, the reversal test (Tauxe, 2010) was employed on polarity zone-mean directions. However, there are some shortcomings with the simple bootstrap of Tauxe's (2010) approach, especially when dealing with a small number of paleomagnetic directions in each polarity mode. First, the number of bootstrapped means for each polarity mode will become the same, while the number of mean directions within each mode are not usually equal in paleomagnetic studies. This will give an artificially higher precision to the polarity mode with a smaller number of directions. Second, in the simple bootstrap reversal test the uncertainty of each site-mean direction is not included, which might bias the outcome toward less-precise site means. Therefore, considering only this issue, the use of the parametric bootstrap reversal test is advised even if a large number of site-mean directions exists for each polarity mode. Third, the distribution of site-mean directions is not necessarily Fisherian. This might get more problematic when dealing with data from time periods of rapid APW, or in cases where an overprint has not been fully removed from the ChRM. The final drawback is lack of robustness with the bootstrap when the number of the data is small, usually ≤3 per polarity mode.

To accommodate small data sets, we modified Tauxe's (2010) approach to perform parametric bootstrap for polarity modes with few (≤3) mean directions. Another modification is using the direction of the

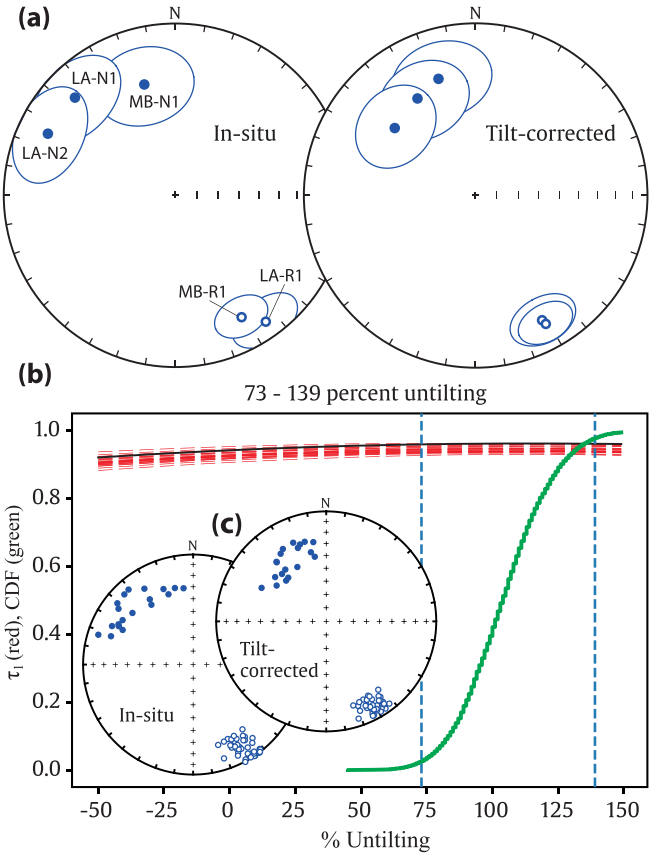


Fig. 9. Fold test results of Summerville Formation zones. (a) Equal-area plots of in-situ and tilt-corrected zone-mean directions. Ellipses are the projection of the cone of 95% confidence about the means. (b) positive bootstrap fold test (Tauxe and Watson, 1994) of simulated mean directions (see text for more explanation). The simulated mean directions are adjusted for incremental tilting from −50% to 150%. (c) Equal-area plots of in-situ and tilt-corrected simulated zone-mean directions. The test is consistent with the magnetization predating tilting because the 95% confidence interval (vertical lines) of the largest eigenvalue (τ_1) encompasses 100% of untilting (73–139%). The number of bootstraps is 10 k.

maximum eigenvector of the site-mean distributions in each polarity mode, as the distribution of the site-mean directions is not necessarily Fisherian. To compare the result of the simple (Tauxe, 2010) and parametric bootstrap reversal test proposed here, the results of the combined Summerville and Morrison Formations' zones means (9 zones) are presented in Fig. 10a and c. This shows the robustness of the parametric bootstrap reversal test, especially where number of the paleomagnetic directions are small. The parametric bootstrap reversal test result of the Summerville Formation zone-mean directions is positive, which is consistent with the magnetization being acquired during or shortly after sedimentation.

For even smaller data sets, e.g., number of site-means in either polarity modes is ≤2, synthetic directions can be created based on the statistical precision parameters of each site mean. Such a simulation was used to perform the simple bootstrap reversal test on the Upper Triassic zones (the only reverse zone is TB-R). Then, the synthetic population of the sample-level directions was fed to the simple bootstrap reversal test (Tauxe, 2010). The result indicates a positive reversal test for the Upper Triassic directions (Fig. 10d).

4.5. Inclination error (IE)

Inclination shallowing or inclination error (IE) in sedimentary rocks results from a tendency for anisotropic magnetic particles to rotate toward horizontal relative to the ambient field while or after being deposited (e.g., Jackson et al., 1991). This effect can be quantified with a flattening factor (*f*), using King's (1955) Eq. (1), where I_C and I_O are true and measured inclinations, respectively:

$$tan\ I_O = f\ tan\ I_C \quad (1)$$

Statistical geomagnetic field models using a great Gaussian process (GGP) (e.g., TK03.GAD (Tauxe and Kent, 2004)) predict the distribution of magnetic directions at different latitudes to be elongated along paleomeridians whereas flattening imparts paleo *E*-W elongation (Tauxe, 2005). The elongation/inclination (E/I) method of Tauxe and Kent (2004) searches for the *f* value that produces the E/I combination compatible with the geomagnetic field model. One of the assumptions of E/I method is abundant (>150) well-defined magnetic directions that have recorded a statistically meaningful paleosecular variation (PSV) (Tauxe et al., 2008).

Paleomagnetic directions obtained as line fits are too few to apply the standard E/I method for all formations separately. Use of site- or zone-mean directions derived from line and GC fits is not ideal for the E/I method because each mean likely averaged a significant portion of the secular variation. To have enough directions to satisfy the assumption of E/I method, we combined several formations. We used the sample-level directions from the line fits of the Upper Triassic Formations ($n = 120$) and the Summerville and Morrison Formations of the Middle Jurassic units ($n = 62$). Presumably, the paleomagnetic directions of the Upper Triassic and Middle Jurassic Formations were acquired over ca. 10 and 5 Myr time spans, respectively. The 5 Myr time span for the Jurassic strata is likely short enough that effects of plate motion on the directional distribution will be negligible, but there could be some effects of motion of North America on the shape-distribution of the Triassic strata if these contain >10 Myr of APW (Beck et al., 2003). Results are presented in Fig. 11 and Table 3. To decrease the influence of polar wander and increase the number of directions, we combined the sample-level directions of this study's Summerville Formation (34 directions) with Steiner's (2003) Trujillo section (62 directions) (see Fig. 11e-f and Table 3 for the E/I results). Steiner's (2003) directions were not tabulated, so were digitized from her Figure 4, which may have introduced some error.

The E/I method results (Table 3) estimate ~19–20° IE for the Middle Jurassic units, although the bootstrap 95% confidence limits are large (~25°). Deenen et al. (2011) introduced reliability criteria for paleomagnetic directions, which should be applied for studies that use geomagnetic field models, like TK03.GAD, to assess IE. The criteria use an N-dependent A95 (95% confidence circles about the paleomagnetic pole) envelope to ascertain that the distribution of the paleomagnetic directions reflects only PSV. These criteria applied to the Jurassic data indicate that all A95's exceed the $A95_{max}$ calculated for the corresponding N. This shows that scatter of the Middle Jurassic directions has other source(s) beside PSV. For our data, the source of this scatter is not known, but could be due to polar wander and not just random noise, and probably denotes excessive time spans considered for the analysis.

The paleomagnetic poles from Upper Triassic strata have reasonable A95s according to the Deenen et al. (2011) criteria, but despite large *f* (0.43) the amount of IE is negligible because the mean direction is horizontal ($I_O = 0.4$; see Table 3).

5. Discussion

5.1. Constraining ages of units

In order to compare the paleomagnetic results from Upper Triassic and Middle Jurassic units studied here to other results from North America or global APWP compilations, the magnetization age should be well defined. Ages for most units are bracketed approximately by unconformity ages (Anderson and Lucas, 1996; Lucas, 2018; Lucas, 2004;

Lucas and Anderson, 1998; Pipiringos and O'Sullivan, 1978), and, for some, by biostratigraphic information in the literature (Imlay, 1980; Lucas, 2010; Wilcox and Currie, 2008). For example, the age of the Redonda Formation is late-Norian to Rhaetian, determined by correlation of the Apachean land-vertebrate faunachron (LVF) (Lucas and Hunt, 1993), which is pinned by the first appearance of a tetrapod genus "*Redondasaurus*" (ca. 215 Ma) (Fig. 12) to the standard global chronostratigraphic scale (Lucas, 2010). The reversal pattern of each unit was then used for correlation with geomagnetic polarity time scales (GPTSs) to determine its magnetization age. The resulting correlation of the sampled units to GPTSs is shown in Fig. 12. The geomagnetic time scales used were Hounslow and Muttoni (2010) for the Late Triassic, and Ogg et al. (2012) for the Middle to Late Jurassic. The Trujillo Formation is dominated by normal polarity zones and can be correlated with UT13n, as the formation occupies Carnian-Norian boundary (Lucas, 1995) (Fig. 12). The Redonda Formation acquired most of its magnetization during a reverse polarity chron, such as after ca. 215 Ma, so is correlated with UT18 (Fig. 12). The Lower-Oxfordian Summerville Formation is correlated with the Callovian-Oxfordian N and early-Oxfordian R zones (Fig. 12).

5.2. Paleomagnetic poles

For the well-sampled formations, paleopoles were calculated (Fig. 13b, and Table 4). In order to find the mean directions for those formations (Dec and Inc. in Table 4), the corresponding zone-mean directions were averaged using PMAG Tools v. 4.2a (Hounslow, 2006). This combines multiple paleomagnetic mean direction using their Fisher (1953) statistics and estimates the overall mean direction and precision parameters as if they had been calculated directly from all directions in the multiple zones. Those mean directions (Table 4) were used to calculated paleomagnetic poles for those well-sampled formations.

To define the paleomagnetic pole of the Middle Jurassic formations, only the zone-means of the reversed polarities were averaged. This is because the next to last component removed (~590° to 660 °C) was generally easier to identify in the reversely magnetized specimens (Fig. 6b, e, and g). The paleomagnetic pole for the Summerville Formation (~163 Ma) from IE-uncorrected directions of reversed zones (LA-R1 and MB-R1, $n = 45$) is 52.4°N, 126.1°E, A95 = 5.4° (Fig. 14). Using $f = 0.46$, estimated with the E/I method, the preferred IE-corrected paleomagnetic pole for 163 Ma is 59.3°N, 138.1°E, A95 = 6.5° (Table 4, Fig. 14b). Using Deenen et al. (2011) formula, the minimum and maximum A95 limits for this pole (n = 45) are 3.7°, and 7.5°, which indicates that PSV has been adequately sampled, while the effect of the unwanted sources of the dispersion is negligible. Although A95 of the Summerville Formation paleomagnetic pole obtained from this combination (A95 = 6.3°) exceeds $A95_{max}$ of Deenen et al. (2011) (Table 3, entry#5), the flattening factor is within range of values expected from this type of rock (Kodama, 2012). More importantly, although a larger number of site directions and/or samples with linear demagnetization behavior might provide better estimation of the flattening factor, no amount for correction for IE would bring Middle Jurassic poles near the high-latitude APWP (Fig. 14a).

Admittedly, E/I analyses for IE were not robust, and the flattening factors (f's) estimated are smaller than the typical f value (0.6) expected for hematite-bearing sediments (e.g., Kodama, 2012) (except for the Morrison Formation), however the f's are plausible. Thus, we employed those f's (Table 3) to correct the formation mean directions and calculated corresponding IE-corrected paleopoles. These poles are, for the Garita Creek and Trujillo Formations (ca. 223 Ma) 55.7°N, 078.7°E, A95 = 3.2°, the Redonda Formation (ca. 214 Ma) at 59.7°N, 081.0°E, A95 = 4.0°, the Summerville Formation (ca. 163 Ma) 59.3°N, 138.1°E, A95 = 6.5°, and the Lower Morrison Formation (Salt Wash Sandstone) (ca. 155 Ma) 48.7°N, 151.8°E, A95 = 7.8° (see Table 4, and Fig. 13b).

5.3. Implication for North American (NA) mid-Jurassic APWP

The high-latitude APWP of NA proposed by Kent and Irving (2010) was derived from igneous rocks from eastern NA; it has a large jump between ~160 Ma and ~145 Ma called the "monster shift". This jump could be due to very rapid movement of the continent relative to the magnetic pole, or true polar wander (TPW) (Kent et al., 2015). However, paleomagnetic results from both igneous and sedimentary units that are more widely distributed define a lower-latitude, more continuous APWP for most of Jurassic time (e.g., Beck and Housen, 2003; Gordon et al., 1984; May and Butler, 1986). These contradictory APWPs have been the

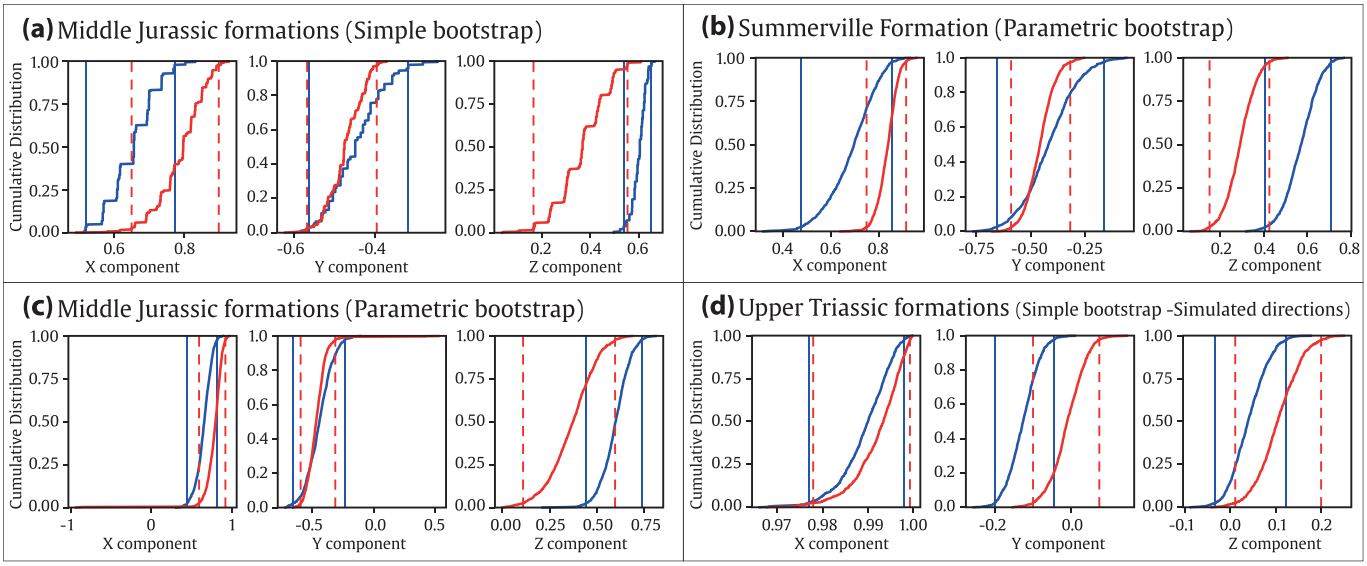


Fig. 10. Bootstrapped reversal test results for (a-c) Middle Jurassic and (d) Upper Triassic strata of east-central New Mexico. (a, and c) are simple (Tauxe et al., 1991) and parametric bootstrap test results for all nine zone-mean directions of Middle Jurassic formations; (b) is the parametric bootstrap test on Summerville Formation zone means; (d) is the result of simple bootstrap test (Tauxe et al., 1991) for Upper Triassic mean directions. (Because there is only one normal zone-mean direction, we simulated sample-level directions based on the statistical precision parameters (k, n) of zone mean directions and used the simple bootstrap reversal test). See text for more explanation. The graphics show the cumulative distribution of directions in Cartesian coordinates, with their 95% confidence intervals (vertical lines). The test is considered to be positive, if the 95% confidence intervals of all X, Y, and Z overlap (Tauxe et al., 1991).

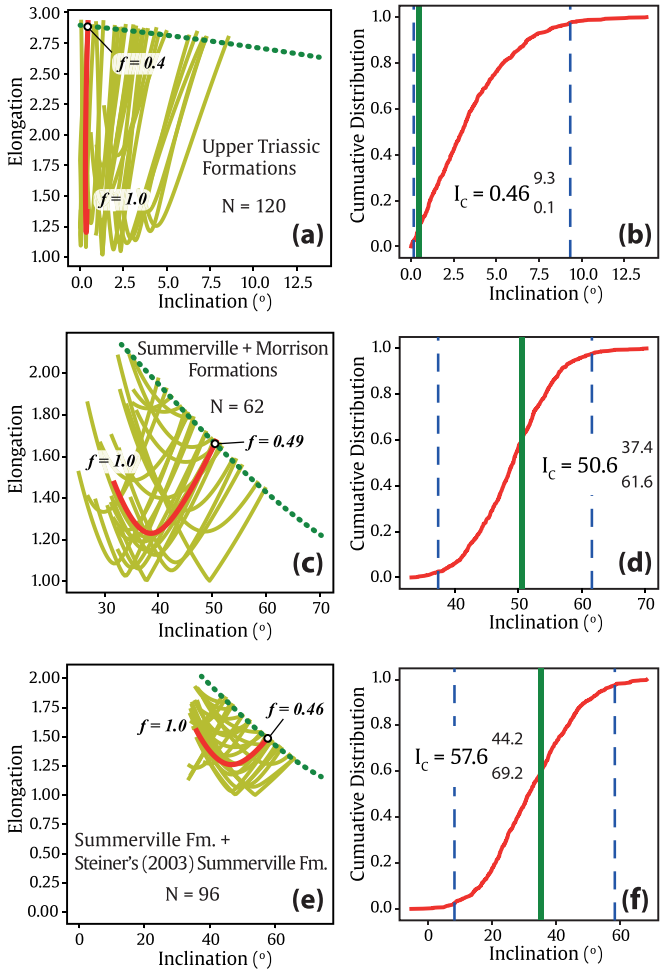


Fig. 11. Elongation/Inclination (E/I) analyses (Tauxe and Kent, 2004) on the sample-level line fits of (a-b) Upper Triassic and (c-d) Middle Jurassic (Summerville and Morrison) Formations of east-central New Mexico. (e-f) This study's Summerville Formation sample-level directions combined with Summerville Formation directions of Steiner (2003) from her Trujillo section. Graphs on the left side show trajectories of mean inclination versus elongation of bootstrapped distributions calculated with values of flattening factors (*f*) ranging from 0.3 to 1.0. The dotted line is the predicted E/I trend of the TK03. GAD geomagnetic field model; in each, the circle is labeled with *f* that corresponds to the bootstrapped E/I mean consistent with the field model. Right-side graphs are cumulative distributions of *f*'s correspond to the 1000 E/I intersections in the left graphs; the dashed vertical lines show the 95% confidence intervals. N: number of directions in each dataset; I_C: corrected mean inclination; +/− I_C: 95% confidence limits for I_C calculated from 1000 bootstraps.

subject of many studies since the 1990s; it is important to reconcile their differences in order to constrain the tectonic evolution of the continent as well as test whether there was TPW.

Among various efforts to resolve this persistent issue, paleomagnetic poles from well-dated Ontario kimberlites (Kent et al., 2015) and Adria (northern Italy) Jurassic and Cretaceous carbonates and radiolarites (Muttoni and Kent, 2019) have been used to support the high-latitude APWP for NA (Kent et al., 2015); the controversy was blamed on inclination error in paleomagnetic results of sedimentary units, most of which are from the U.S. Southwest interior (Kent and Irving, 2010). As previously mentioned, and very nicely summarized in Muttoni and Kent, 2019, there is a significant discrepancy in the pole positions of mid-upper Jurassic sedimentary rocks (Summerville and Morrison Formations) and the pole positions that define two polar still-stands (one from 190 to 160 Ma, another from 140 to 60 Ma), with an interval of rapid polar motion (their proposed TPW and/or Monster Shift) between 160 and 145 Ma. Results from ~156 Ma kimberlites from Ontario are proposed to capture a portion of this rapid polar wander. These NA APWPs have widest separation in the middle Jurassic, which corresponds to the age of the Summerville Formation. Therefore, the choice between the two paths depends on the accuracy, both in age and direction, of the Summerville results compared to those from the Ontario kimberlites and Adria. Below are some thoughts about strengths and weaknesses of each.

5.3.1. *Results from southwestern of North America*

The zone-mean directions of the Summerville Formation are superposed on the paleomagnetic direction trajectories predicted from the controversial APWPs for east-central New Mexico coordinates (Fig. 15). The zone-mean directions, specifically those from reverse zones, are in complete agreement with the trend calculated from the Beck and Housen (2003) path. This is consistent with APWP for North America proposed by other compilations that are derived mostly from sedimentary rocks (e.g., Beck and Housen, 2003; Gordon et al., 1984; May and Butler, 1986). Similarity of those paths confirms that results from these rocks can be successfully replicated. Although a95 confidence cones (semi-angle of the 95% confidence about the mean) for some zone-directions (e.g., MB-N1) encompasses both trends, that could be due to the small number of directions used to define the means (n = 4, see Table 2) and/or an unrecognized steeper normal overprint. Moreover, the Summerville Formation reverse-zone means cluster around the 170–160 Ma sector of the predicted directional trend (Fig. 15), which is consistent with the suggested mean magnetization age (ca. 163 Ma) for this formation (Table 4). This also strengthens the conclusion that the ChRM component is likely of primary origin (magnetization age at ~163 Ma) as indicated by the fold and reversal tests results of the Summerville Formation (Figs. 9 and 10b).

Most of the rock units studied were deposited in low accumulation rate environments and most of our zones encompasses several meters of sections (e.g., in zone TB-R, thickness = 27 m, and $n = 78$, see Tables 1 and 2). Therefore, PSV might have been averaged in the sample-level data such that zone VGPs would be good approximations of the paleopole for those zones. This would also imply that the VGPs' A95s should be used for poles in Fig. 13a. The VGPs of the zones (excluding ones from odd directions and zone H01-TA-R) are shown in Fig. 13a (also see Table 2). Calculation of most zone-mean directions was by combining line and GC fits, so calculation of A95s for zone VGPs was not possible by simple transformation of the sample directions to pole coordinates. Instead, we used the precision parameters of the zone-mean directions and their paleolatitudes (calculated using the Dipole Formula: tan [magnetic inclination] = 2 tan[latitude]) in Cox's (1970) method to determine corresponding A95s (Fig. 13a). All of the Late Triassic poles are perfectly streaked along the 245–200 track of the lower-latitude APWP, and the Middle Jurassic poles hover around the 160 Ma cusp of the same path (Fig. 13a).

These results for the Lower Morrison Formation from New Mexico differ from those obtained by Muttoni and Kent (2019) at the Norwood location. They lack a coherent overprint, such as found in the Norwood locality (Muttoni and Kent, 2019) (Fig. 7). Interestingly, such remagnetization is also absent in Upper Triassic redbeds on the Colorado Plateau (Kent et al., 2018, 2019). Moreover, detailed IRM-analysis of representative specimens from the lower Morrison Formation that we studied (Fig. 6g) revealed four magnetic phases based on coercivity spectra. One, whose $B_{1/2}$ values (component 4 in Fig. 6g) is very high, is likely an iron-hydroxide. Two phases with high coercivities (also confirmed by the demagnetization plot), one of which ($B_{1/2}$ = 216 and contribution = ~37%) likely is hematite with relatively large grain size, which would have very high T_{ub} with very narrow ranges (Jiang et al., 2015; Özdemir and Dunlop, 2014; Swanson-Hysell et al., 2019; Tauxe and Kent, 1984). The demagnetization plots also confirm that two phases of hematite with relatively high T_{ub} ranges exist, one unblocked between ~600 °C and 650 °C and the other unblocked at >660C

Table 3
Results of E/I analysis (Tauxe and Kent, 2004) of sample-level directions from line fits of Upper Triassic and Middle Jurassic formations of east-central New Mexico. A95 (95% confidence circle about the paleomagnetic pole) of all Jurassic formations are slightly higher than Deenen et al. (2011) criteria, which indicates that the dispersion of VGPs has sources other than paleosecular variation (PSV) (see text for further explanation).

Formation	N	D (°)	I_O (°)	*f*	I_C (°)	+/− I_C (°)	Plat′	Plong′	A95′	Plat	Plong	A95	Deenen et al. (2011) *criteria*		
													$A95_{max}$	$A95_{min}$	*A95 criteria*
Upper Triassic units	120	359.1	00.40	0.43	00.46	0.1–9.3	55.6	076.7	4.9	55.3	076.9	4.0	*4.0*	*1.8*	*Ok*
Summerville Fm.	34	329.1	28.45	0.44	49.06	33.3–62.5	63.7	163.6	9.8	56.6	137.6	9.4	*8.9*	*2.9*	*High*
Morrison Fm.	28	314.0	36.20	0.98	36.72	32.8–66.7	47.4	160.6	10.9	47.2	160.1	10.9	*10.0*	*3.2*	*High*
Summerville Fm. & Morrison Fm.	62	322.6	31.80	0.49	50.65	37.4–61.6	59.1	170.6	7.3	52.9	148.9	7.3	*6.1*	*2.3*	*High*
Summerville Fm. & Summerville Fm. *	96	324.5	35.96	0.46	57.61	44.2–69.2	61.8	183.3	6.3	56.1	152.3	6.6	*4.6*	*1.9*	*High*

*: Sample-level Summerville Formation directions of Steiner (2003) from her Trujillo section; N: number of directions in each dataset (normal and reversed zones); D and I_O: declination and inclination of the observed mean direction; *f*: flattening factor determined from E/I analysis; I_C: corrected mean inclination; +/− I_C: 95% confidence limits for I_C calculated from 1000 bootstraps; Plat, Plong, and A95 (Plat′, Plong′, and A95′): coordinate of the paleopole and radius of its 95% confidence circle calculated from the distribution of VGPs (IE-corrected directions transformed to VGPs); $A95_{min}$ and $A95_{max}$: upper and lower limits of A95, which provide reliability criteria to examine the role of PSV in dispersion of the paleomagnetic VGPs (Deenen et al., 2011).

(Fig. 6g). The small temperature steps we employed above ~650 °C enabled us to capture this very high T_{ub} component. This could not have been done using the fixed steps of 650°, 660°, 670° and 675 °C that Muttoni and Kent (2019) used for all specimens. However, their Figure 4a (specimen JMB11a) shows that these two components also exist in their samples, one between 600 °C and 660 °C and the other >660 °C, the latter being shallower and more southeasterly. It may be possible that the ChRM of Muttoni and Kent (2019) if revised to use the highest temperature component, could change the reversed directions to be more antipodal, such that their data may then have passed the reversal test. Regardless, our results do suggest that the steeply-inclined B component overprint, and highly scattered ChRM directions Muttoni and Kent (2019) obtained from the Morrison Formation are not present in the Summerville (and Morrison) Formation(s) from east-central New Mexico.

Steiner (2003) previously sampled and studied the cratonic Summerville Formation. Those samples were taken from the Romeroville Gap (this study's Las Vegas section) and the Trujillo Mound sections (locations L and T in Fig. 1). In the former locality, only the first ~10 m mudstone-dominated section was interpreted and reported as the Summerville Formation (Steiner, 2003), and the rest of the section was assigned to the Lower and Upper Morrison Formation (Steiner et al., 1994). However, more recent paleontology and stratigraphy studies showed that the lower ~50 m part of the Las Vegas section, which is pale-red fine sandstone and mudstone, and which Steiner (2003) interpreted as the Lower Morrison Formation, is actually Upper Summerville strata and is homotaxial with the Bluff Sandstone of San Rafael Group (Lucas, 2018; Lucas et al., 1999; Lucas and Woodward, 2001). To increase the number of directions to meet Tauxe et al. (2008) assumption for E/I method, we combined the sample-level directions of this study's Summerville Formation ($n = 34$) with Steiner's (2003) Trujillo section ($n = 62$) obtained from digitizing her Figure 4 (see Fig. 11e-f and Table 3 for the E/I results). The resulting paleomagnetic poles are plotted in Fig. 14a, as pole position 1 uncorrected for IE, and position 2 IE-corrected with $f = 0.46$ (Table 3).

To further illustrate possible effects of inclination flattening, we used three reported Summerville Formation paleomagnetic poles to simulate the position of their corresponding IE-corrected poles (Fig. 14a). Two are from cratonic parts of NA (this study [52.4°N, 126.1°E], and Steiner (2003) [56.9°N, 143.3°E]) and one from the Colorado Plateau reported by Bazard and Butler (1992) corrected for 10.5° clockwise rotation of the plateau estimated by Beck and Housen (2003) [48.8°N, 142.7°E]. IE-corrected inclinations were calculated using King's (1955) equation and a range of *f* values (0.1–0.9), and then the new sets of declination and inclination were transformed to pole coordinates (Fig. 14a). From Fig. 14a it is evident that no amount the IE correction of the sedimentary rocks from the SW portion of NA would bring the two contradictory APWPs into an agreement.

The results here from the Summerville and portions of the lower Morrison Formation, while possessing some scatter, appear to have recorded the geomagnetic field with high fidelity. The sections sampled have numerous and distinct polarity zones, directions from which pass the paleomagnetic reversals test, and are better clustered after correction for bedding tilt. These analyses indicate that a ChRM direction that likely dates from deposition of these units has been isolated. The polarity zones and paleomagnetic poles obtained from this study are comparable to other results from the Summerville Formation (Bazard and Butler, 1992, and Steiner, 2003), thus the paleomagnetic results from the Summerville Formation have been successfully reproduced by different workers over a span of several decades.

5.3.2. *Results from eastern North America and Europe*

Paleomagnetic poles used to support the high latitude path for middle Jurassic, come from Jurassic-Cretaceous carbonates and radiolarites of Adria from Bombatierle and Foza sections in the central eastern Southern Alps (Muttoni and Kent, 2019). Their magnetostratigraphy and nannofossil content were extensively studied by Channell et al. (2010). The IE-corrected paleomagnetic poles from these sections (rotated to the North American coordinates are shown in Fig. 14b as 158Bomb for Bombatierle, and 158Foz for Foza sections. (see Muttoni and Kent (2019), (Table 1)) for more information about rotation parameters). These rocks appear to have captured the rapid polar shift along the high-latitude APWP (Muttoni and Kent, 2019). However, both Bombatierle and Foza sections lack age-diagnostic calcareous nannofossils for the middle Jurassic portions of the section, so no zonal or age assignments could be made (Channell et al., 2010). Moreover, due to the poor definition of reverse polarity chrons, magnetostratigraphic correlations with the GPTS were not possible before ~150 Ma, i.e., below CM22 at Foza and CM20 at Bombatierle (Channell et al., 2010; Muttoni and Kent, 2019). So, the nominal age of 158 Ma assigned to these poles (Muttoni and Kent, 2019) is merely based on lithostratigraphic correlation between the Bombatierle and Foza sections and the Rosso Ammonitico Medio. The ash layer at the top of the latter has been dated at 156.1 +/− 0.89 Ma (Pellenard et al., 2013; Ar/Ar). Thus, the magnetization could be older than the assigned nominal ages of 158 Ma. Furthermore, the apparent separation of 158Bomb and 158Foz poles might arise from incorrect age assignment rather than rapid polar motion. Additional uncertainty could be due to vertical-axis rotations of sampled sections in Adria attributed to Umbrian deformation (see Satolli et al., 2007, and Channell et al., 2010, for more details).

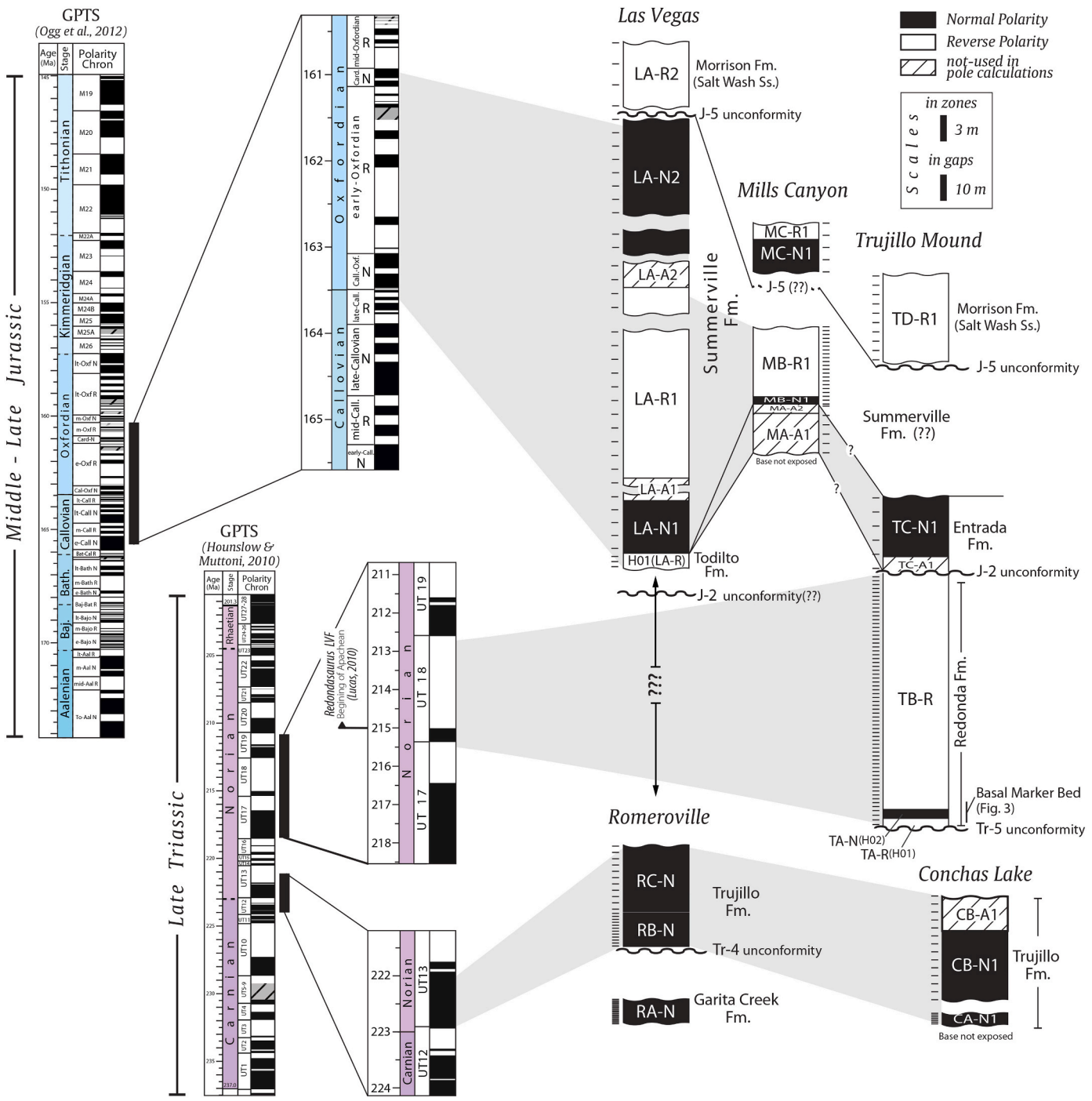


Fig. 12. Correlation of magnetic polarity patterns of sections studied here to geomagnetic polarity time scale (GPTS), used to estimate ages of these units; sources are at tops of columns. Tic marks along the columns are approximate sampled horizons. Shading between time scales on left and magnetostratigraphic columns on right show correlations based on ages available for the stratigraphic units and polarity observed. Note that vertical scale on the left is in time, on right is in distance but different for sampled intervals and gaps between.

5.3.3. *Resolution*

Data and analysis presented here indicate that even with IE-correction the North American APWP would streak approximately along the 60° parallel (in current coordinates) for most of the Jurassic time, and the lingering APWP discord is not attributable to IE of the sedimentary units.

In contrast with results from southwest North America, the high-latitude APWP proposed by Kent and Irving (2010) has relatively few poles for Mid- to Late- Jurassic time. The 160 Ma mean pole is based on four poles; only one of which, the Moat volcanics (Van Fossen and Kent, 1990), is from NA. Those data come from a collapsed caldera, so paleohorizontal control may perhaps be suspect. Paleomagnetic results (sample VGPs) from radiometrically-dated Triple B (ca. 154.9 Ma) and Peddie (ca. 157.5 Ma) kimberlites established a mean paleomagnetic pole (75.5°N, 189.5°E, A95 = 2.8°, n = 34, 156Ok in Fig. 14b) (Kent et al., 2015), to which Muttoni and Kent (2019) assigned an average age of 156 Ma (Fig. 14b). Besides the lack of paleohorizontal control in these kimberlites, the pole's A95 is less than the lower A95 limit of 2.9° calculated via method of Deenen et al. (2011), which shows that PSV may not have been adequately sampled.

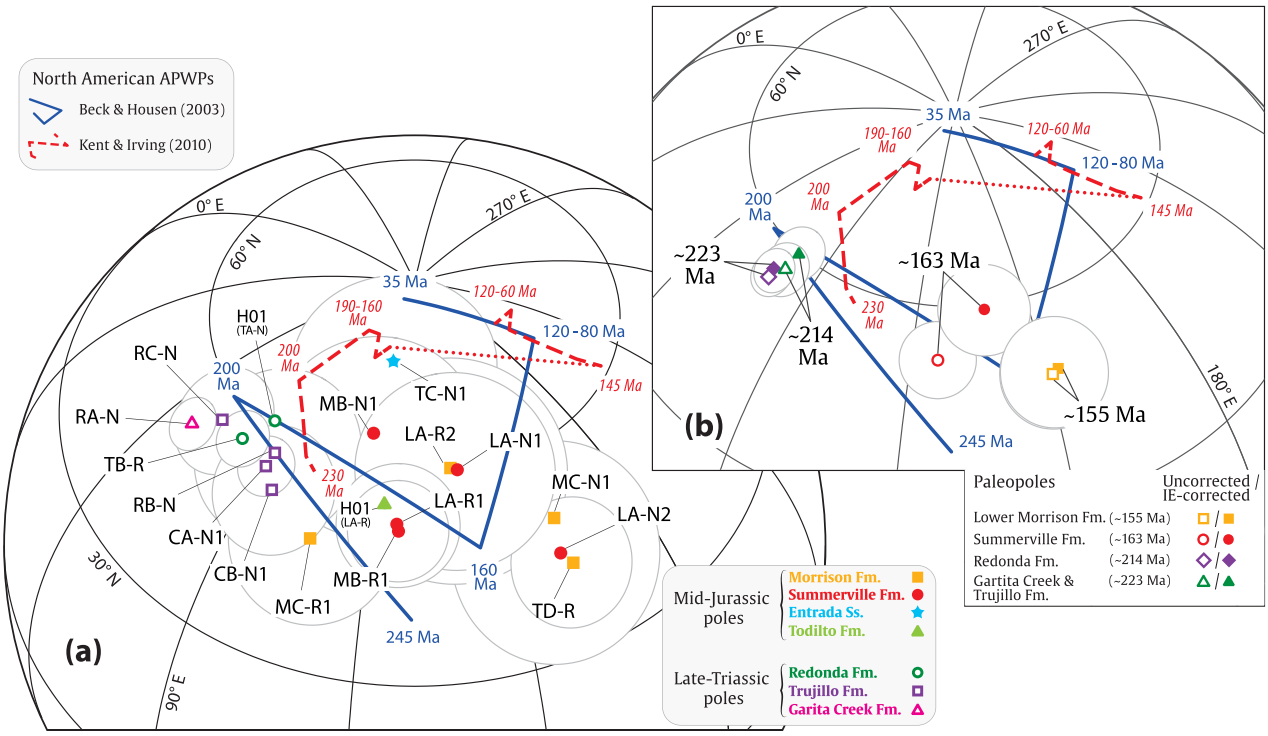


Fig. 13. (a) Virtual geomagnetic poles (VGPs) of Upper Triassic and Middle Jurassic zones of east-central New Mexico. (b) Uncorrected (open) and IE-corrected (solid) paleomagnetic poles of well-sampled formations of the study area (see Table 4 for more information). Dotted part on Kent and Irving (2010) APW path marks the "monster shift". Circles are A95 confidence limits about the poles (>20° are not drawn), which are indirectly calculated using Cox's (1970) method.

Table 4
Paleomagnetic pole coordinates of Upper Triassic and Middle Jurassic units of east-central New Mexico for the well-sampled formations of this study.

Formation	Age (Ma)*	Mean unit coordinates		Mean direction				*f*	Inc′	Pole coordinates					
		Lat.	Long.	N	Dec	Inc	a95			Plat.	Plong.	A95	Plat′	Plong′	A95′
Middle Jurassic															
Morrison (Salt Wash Mbr)	155	35.573	255.261	30	137.5	−32.8	10.1	0.98	−33.3	48.5	151.4	7.7	48.7	151.8	7.8
Summerville	163	35.733	255.234	45	151.4	−16.9	8.4	0.46	−33.4	52.4	126.1	5.4	59.3	138.1	6.5
Upper Triassic															
Redonda	214	35.514	255.316	81	177.1	−4.6	6.5	0.43	−10.6	56.7	080.6	3.9	59.7	081.0	4.0
Garita Creek and Trujillo	223	35.453	255.035	75	357.9	01.1	5.2	0.43	02.6	55.0	078.7	3.2	55.7	078.7	3.2

*: Age is the mean age inferred from both Fig. 14 and the literature (Pipiringos and O'Sullivan (1978) for the Salt Wash Member); N: sum of the sample directions from all averaged zones (reversed zones are averaged for the middle Jurassic units); Dec and Inc.: declination and inclination of the mean direction; *f*: flattening factor determined from E/I analysis (see Table 3); Inc′: corrected mean inclination; Plat, Plong, and A95 (Plat′, Plong′, and A95′): coordinates of the paleomagnetic pole and radius of its 95% confidence circle indirectly calculated using Cox's (1970)(same but from IE-corrected means).

Therefore, it is suggested that the high-latitude APWP (Kent et al., 2015; Kent and Irving, 2010; Muttoni and Kent, 2019) may suffer from inadequately-sampled PSV, age uncertainty, and lack of rigorous paleohorizontal control, although we think that the latter should be the prime suspect. Poles for other ages along the high latitude path derived from eastern NA might also suffer from regional tilt due to vertical movement of the continental margin as a consequence of post-rifting thermal uplift/subsidence.

6. Conclusions

Paleomagnetic results of Upper Triassic (Garita Creek, Trujillo, and Redonda Formations) and Middle Jurassic (Entrada, Summerville, and Lower Morrison Formations) sedimentary units of east-central New Mexico have magnetostratigraphy that can be correlated to the GPTS, although more detailed work is required to achieve better-constrained magnetization ages. Most of the sampled units are fine-grained sandstones, siltstones, and mudstones. Rock-magnetic analysis indicates that hematite (mostly detrital) is the main remanence carrier.

In a general sense, remanent magnetism and consequently the paleomagnetic data are less noisy and scattered from the Upper Triassic units than from the Middle Jurassic ones. The lower quality of the Middle Jurassic data could be due to high rate of polarity reversals for this time period (Biggin et al., 2008). Although noisier than the Triassic data, the paleomagnetic data from Middle Jurassic units (Summerville and Lower Morrison Formations) appear to record the geomagnetic field with high fidelity. The sections sampled have numerous and distinct polarity zones; directions from these, pass the paleomagnetic reversals test, and have better clustering of directions after correction for bedding tilt. These results are consistent with the ChRM isolated likely dating from deposition of these units.

Most of these units were deposited in low accumulation-rate environments, and most of our zones encompasses several meters of sections, so the PSV might have been averaged within zones. Therefore, a zone-VGP might be a good approximation of the paleopole for the corresponding zone. Paleomagnetic directions from zones within units are combined to define formation-mean directions and paleopoles for Trujillo, Redonda, Summerville, and Lower Morrison (Table 4).

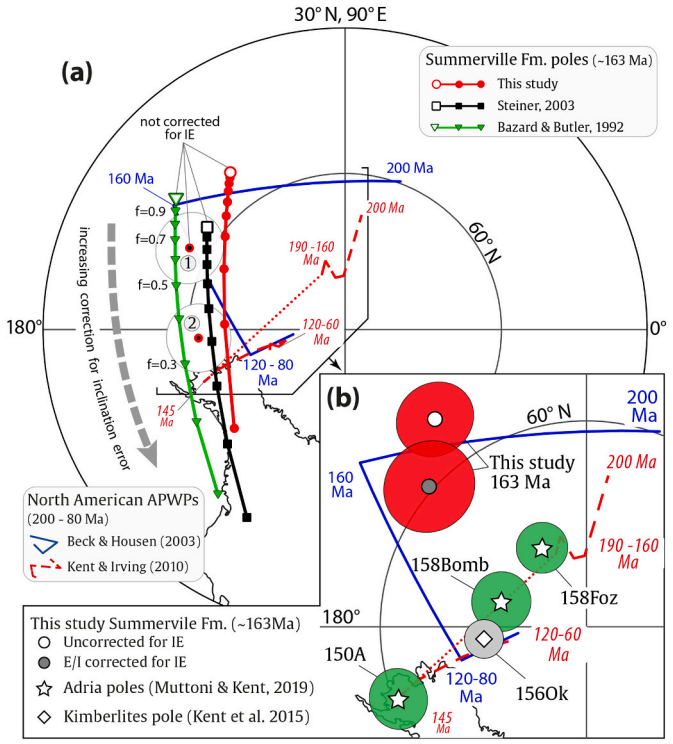


Fig. 14. (a) Paleomagnetic poles of Summerville Formation and their simulated paths for inclination error (IE) corrections using King's (1955) equation and a range of flattening factor (f) of 0.1 to 0.9. Triangle is Bazard and Butler (1992) pass C sites of Colorado Plateau's Summerville Formation; square is the group I cratonic pole of Steiner (2003), and circle is this study's preferred pole calculated using directions from reversed zones (LA-R1, and MBR1). 1: Combined Summerville Formation paleopole, calculated from sample-level line fits of this study (n = 34 [N + R]) and all sample-level directions of Steiner's (2003) Trujillo section (n = 62); 2: Pole position of (1) corrected for inclination error (IE) using E/I method of Tauxe and Kent (2004). Dotted part on Kent and Irving (2010) APW path marks the "monster shift". See Fig. 10 and Table 3 for more information. This shows that IE correction cannot bring two conflicting APWPs into agreement as suggested by some researchers (e.g., Kent and Irving, 2010); b) Preferred ~163 Ma paleopole from Summerville Formation (open/closed circle: uncorrected/corrected [f = 0.46]) of this study (derived from reversed zones: LA-R1, and MBR1). Stars: Adria poles (Muttoni and Kent, 2019) for ~158 Ma (158Bomb, and 158Foz), and ~ 150 Ma (entry #15 of Table1 of Muttoni and Kent, 2015). Diamond: the ~156 Ma Ontario kimberlites pole of Kent et al. (2015). See text for more information and discussions.

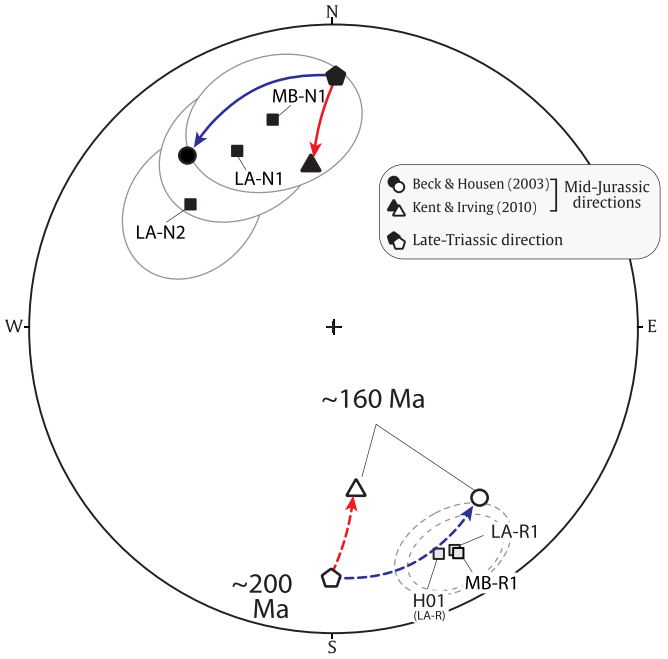


Fig. 15. Equal-area plot of change in directions during late Triassic – Middle Jurassic calculated from two contradictory apparent polar wander paths (APWP) of North America for mean coordinates of Middle Jurassic Formations of this study (35.733 N, 255.234 E). The Summerville Formation zone-mean directions (black and gray-filled squares for normal and reversed zones, respectively) plot on the Beck and Housen (2003) expected trend, which validates the lower latitude APWP for North America (e.g., Gordon et al., 1984; May and Butler, 1986), and also confirms a magnetization age of ~163 Ma for this Formation.

Use of great-circle fits to demagnetization data, mostly from the Middle Jurassic units, reduced the number of sample-level directions to less than the number of samples, so the number that could be used to evaluate inclination error (IE) using the elongation/inclination (E/I) method of Tauxe and Kent (2004) was less than ideal. To improve estimation of f, this study' sample-level line fits for the Summerville Formation were combined with ones obtained from Steiner (2003). An estimated flattening factor (f = 0.46) was used to correct the paleomagnetic directions (LA-R1 and MB-R1, n = 45). Those IE-corrected directions yielded a 163 Ma paleopole at 59.3°N, 138.1°E, and A95 = 6.5°. Using criteria of Deenen et al. (2011), this A95 indicates that the PSV was adequately averaged without extraneous sources of scatter. The Upper Triassic units, despite large f values, require almost no correction for IE because the mean direction is horizontal (I_O = 0.4).

The difference between the corrected Summerville pole, and paleomagnetic poles presented by Kent et al. (2015, Ontario kimberlites, 156 Ma) and Muttoni and Kent (2019, imported poles from Adria, 158 Ma), for 160–150 Ma age range is large. If the high-latitude Monster Shift model is correct, then the Summerville Formation would need to have a magnetization that recorded the younger end of this event, and would need an unrealistic amount of IE (Fig. 14b). This is unlikely considering that evidence here indicates that the magnetization was acquired during deposition or soon after that.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Reviews by Dr. K. Kodama and Dr. R. Molina Garza are greatly appreciated for their help in improving clarity of this manuscript. This study was funded by GSA Research Grant (2017), and internal grants from Western Washington University. Special thanks to Dr. S. G. Lucas for his guidance to better age-determination of the sampled sections, Dr. K. Zeigler for her consultations that allowed us to identify the desired sampling section, and William Callebert for his assistance with paleomagnetic sampling.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tecto.2021.228872.

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